

Combined Reference Texts

Non-European Mathematics Corpus - current checked work-level components

Reference texts from the current checked work-level PDFs.

Components

1. 70-01 Reference Text - Said al-Andalusi - Tabaqat al-Umam.pdf (15 pages)
2. 70-02 Reference Text - al-Muqaddasi - Ahsan al-Ta'asim.pdf (10 pages)
3. 70-03 Reference Text - Ibn al-Nadim - Kitab al-Fihrist.pdf (11 pages)
4. 70-04 Reference Text - Ibn al-Qifti - Tarikh al-Hukama.pdf (10 pages)
5. 70-05 Reference Text - Smith-Karpinski - Numerals.pdf (23 pages)

كِتَابُ الْأُمَمِ طَبَقَاتِ

الْأَنْدَلُسِيُّ صَاعِدِ بْنِ أَحْمَدَ بْنِ صَاعِدِ الْقَاسِمِ أَبِي الْقَاضِي

(م ١٠٧٠-١٠٦٩ هـ ٤٦٢ سنة المتوفى)

وَالْفَهَارِسِ بِالرَّوَايَاتِ وَارِدَةً بِالْحَوَاشِي وَذِيْلُهُ نَشْرُهُ

الْيَسُوعِيُّ شَيْخُ لُؤَيْسِ الْأَبِّ

الشَّرْقِ جَمَلَةً مِنْ عَشْرَةِ الرَّابِعَةِ السَّنَةِ فِي بَتْنَابِجِ نَشْرِهِ

الْيَسُوعِيِّينَ لِلْأَبَاءِ الْكَاثُولِيكِيَّةِ الْمَطْبَعَةُ

KITAB TABAQAT AL-UMAM

(BOOK OF THE CATEGORIES OF NATIONS)

By

**Abu al-Qasim Sa'id ibn Ahmad ibn Sa'id
al-Andalusi**

(d. 462 AH / 1070 CE)

Edited with Notes, Variants, and Indexes by

Pere Louis Cheikho, S.J.

Al-Matba'a al-Kathulikiyya li-l-Aba' al-Yasusiyyin

BEIRUT, 1912

Reference-text reader

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Editor's Introduction

الْأُمَمِ طَبَقَاتِ كِتَابُ
الْأَنْدَلُسِيِّ صَاعِدِ بْنِ أَحْمَدَ بْنِ صَاعِدِ الْقَاسِمِ أَبِي الْقَاضِي

تَوْطئة

سَبَقَتْ الَّتِي الْأُمَمِ بَيْنَ الْعُلُومِ لَوْصِفِ الْعَرَبِ كَتَبَهُ فِيهَا نَعَرَضُ الَّتِي النَّادِرَةِ الْكُتُبِ أَحَدُ الْأُمَمِ طَبَقَاتِ كِتَابُ
عَلَى تَدُلُّ فَوَائِدَ عِدَّةَ جَمَعَ أَنَّهُ إِلَّا الدَّمِيمِ ابْنُ الْفَرَجِ لِأَبِي الْفَهْرَسْتِ كِتَابُ شَاوُ ذَلِكَ فِي صَاحِبِهِ يَبْلُغُ لَمْ وَإِنْ عَهْدَهُمْ،
لَا هِلَ وَبِرُّوْسٍ بِهِ يَفْتَحِرُونَ الْأَنْدَلُسِ أَهْلُ وَكَانَ التَّنَوُّنِ، فِي نَظَرٍ وَدَقَّةٍ التَّحْصِيلِ فِي رَغْبَةٍ وَعَلَى الْبَحْثِ فِي نَشَاطٍ
مَرْزُوقِ بْنِ مُحَمَّدِ بْنِ اللَّهِ عَبْدِ عَنْ بَجَرِيْطَ) طَبْعَةُ ٤٦٣ : (٢ الصَّلَةِ لِكِتَابِ التَّجَلَّةِ كِتَابُ فِي الْأَبَارِ ابْنُ ذَكَرٍ وَقَدْ الشَّرْقِ.
السَّلَفِيِّ. طَاهِرٍ لِأَبِي الْكِتَابِ هَذَا رَوَى الْإِسْكَندَرِيَّةُ قَدَّمَ لِمَا اللَّهُ الْيَحْصِي

الدُّوَلِ مُخْتَصَرِ تَارِيخِ كِتَابِ فِي عَنْهُ نَقَلَ فَإِنَّهُ الْعَبْرِيِّ ابْنُ غَرْيُورِيُوسَ الْفَرَجِ أَبُو الشَّرْقِ فِي الْكِتَابِ هَذَا عَرَفُوا وَمَنْ
فَذَكَرَهُ خَلِيفَةُ الْحَاجِّ عَرَفَهُ وَكَذَلِكَ وَعُلُومِهِمْ، الْعَرَبِ فِي مُفِيدَتَيْنِ نَبَذَتَيْنِ الْبِيْرُوتِيَّةِ) طَبْعَتَا مِنْ ١٣٥ و ١٥٨ (ص
إِنَّهُ وَصَفَهُ فِي وَقَالَ الْأُمَمِ بِطَبَقَاتِ التَّعْرِيفِ لِبَسِيْكَ) طَبْعَةُ ٨١٢ : ٣ (فِي تَارَةً فَدَعَا الظُّنُونِ كَشَفَ كِتَابِهِ فِي مَرَارًا
اعْتَبَارٍ عَلَى دَلِيلًا الْمُنْقُولَاتِ هَذِهِ وَكَفَى الْأُمَمِ. طَبَقَاتِ كِتَابُ ١٣٣) : (٤ وَتَارَةً النَّفْعِ، كَثِيرُ الْحُجْمِ صَغِيرُ كِتَابُ
وَمَوْثُوقُهُ. لِلْكِتَابِ الْقُدَمَاءِ

[Translation of the Editor's Introduction]

Kitab Tabaqat al-Umam (The Book of the Categories of Nations) is one of the rare works in which an Arab writer describes the sciences among the nations that preceded his era. Although its author did not reach the level of Abu al-Faraj ibn al-Damimi in his *Fihrist*, he nevertheless collected numerous useful facts that demonstrate his research activity, his desire for knowledge, and his precision in observation. The people of al-Andalus used to boast of him and of the heads of the East.

Ibn al-Abb'ar mentions in his *Takmila* to the *Silah* (2:463, Bajarid edition) on the authority of Abd Al-lah ibn Muhammad ibn Marzuq al-Yahsibi—may God reward him for what he presented to Alexandria—that he transmitted this book to Abu Tahir al-Salafi.

Among those who knew this book in the East was Abu al-Faraj Grigorius ibn al-'Ibri, for he quoted from it in his *History of the Dynasties* (p. 158 and 135 in our Beirut edition) two useful excerpts concerning the Arabs and their sciences. Likewise, al-Hajj Khalifa knew it and mentioned it several times in his *Kashf al-Zunun*: at one point (3:812, Lipsik edition) calling it *Ta'rif bi-Tabaqat al-Umam*, and describing it as a small book of great benefit; at another point (4:133) as *Kitab Tabaqat al-Umam*. These quotations suffice as proof of the esteem and trust the ancients had for this book.

وَرَقٍ بِجِلْدٍ مُتَقَنَّاً تَجْلِيدًا وَمَجْلَدَةً الصَّفْرَةَ إِلَى ضَارِبِ صَفِيْقٍ وَرَقٍ عَلَى الْفَارِسِيِّ بِالْقَلَمِ شَبِيهِ جَلِيٍّ بِخَطِّ مَكْتُوبَةٍ وَهِيَ
أَمْكَنَّا الَّتِي الْأَغْلَاطِ مِنْ فِيهَا وَقَعَ مَا مَعَ حَسَنَةً بِالْإِجْمَالِ وَالنُّسخَةُ زِينَةً، مِثْلَهُمَا لِسَانٍ مَعَ الْوَجْهَيْنِ عَلَى ذَهَبِيٍّ مُلَوَّنِ
طَبْعَتْنَا. ذَيْلٌ فِي عَلَيْهَا فَتَبَيَّنَ أَكْثَرُهَا إِصْلَاحُ

ص بَجَرِيْطَ (طَبْعَةُ الصَّلَةِ كِتَابٍ فِي بَشْكَوَالِ ابْنِ رَوَاهَا كَمَا تَرَجَّمَتْهُ وَهَذِهِ أَمْرُهُ، مِنْ الْأَقْلَّ نَعْلَمُ فَلَا الْمُوَلَّفُ أَمَّا عَنْهُ: قَالَ ٢٣٤)

عَنْ رُوِي قُرْبَةَ مِنْ وَأَصْلُهُ الْقَاسِمُ أَبِي بَكْنِيَةِ الْقَاضِي التَّغْلِي صَاعِدِ بْنِ مُحَمَّدِ بْنِ الرَّحْمَنِ عَبْدِ بْنِ أَحْمَدَ بْنِ >>صَاعِدِ
وَكَانَ بَطْلَيْطَلَةَ التُّونِ ذِي بْنِ يَحْيَى الْأَمِيرُ وَاسْتَقْضَاهُ وَغَيْرَهُمْ، الْوَقْشِيُّ الْوَلِيدُ وَأَبِي قَاسِمِ بْنِ وَالْفَتْحِ حَزْمِ بْنِ مُحَمَّدِ أَبِي
نَظَرِهِ، أَيَّامَ بِذَلِكَ وَقُضِيَ الْخُطُّ عَلَى وَبِالشَّهَادَةِ الْحَقُّوقِ فِي الْوَاحِدِ الشَّاهِدِ مَعَ الْيَمِينِ الْقَضَاةَ وَاخْتِبَارَ أُمُورِهِ فِي مُتَحَرِّراً
فِي قَاضِيَا وَهُوَ بَطْلَيْطَلَةَ وَتُوِّفِيَ (م) ١٠٣٩ = (هـ) ٤٢٠ سَنَةً فِي بِالْمَرْيَةِ وَلِدَ وَالرَّوَايَةِ، وَالذِّكَاةُ الْمَعْرِفَةُ أَهْلِي مِنْ وَكَانَ
ابْنُ بَعْضُهُ ذَكَرَ الْحَدِيدِيِّ. بْنِ سَعِيدِ بْنِ يَحْيَى عَلَيْهِ وَصَلَّى (م) ١٠٧٠ = هـ (٤٦٢ مِائَةً وَأَرْبَعٍ وَسِتِّينَ اثْنَتَيْنِ سَنَةً شَوَّالِ
مُطَاهِرٍ >>

[Manuscript Description]

[The manuscript is written in a clear hand resembling Persian script, on thick paper tending toward yellow, and bound with skillful binding in colored gold paper on both sides with a matching tongue as ornament. The manuscript is, in general, fine despite the errors that occurred in it, most of which we were able to correct and have indicated in the notes to our edition.]

[Biography of the Author]

As for the author, we know no less than what Ibn Bashkuwal narrated in his *Silah* (Bajarid edition, p. 234):

“Sa'id ibn Ahmad ibn Abd al-Rahman ibn Muhammad ibn Sa'id al-Taghlabi, the judge with the kunya Abu al-Qasim, originally from Cordoba, transmitted from Abu Muhammad ibn Hazm, al-Fath ibn Qasim, Abu al-Walid al-Waqshi, and others. The amir Yahya ibn Dhi al-Nun appointed him as judge of Tulaytula (Toledo), and he was independent in his affairs and in testing judges by oath along with one witness in lawsuits and by testimony regarding handwriting, and judgments were made accordingly during his tenure. He was a man of knowledge, intelligence, and transmission. He was born in al-Mariyya (Almeria) in 420 AH (1039 CE) and died in Tulaytula while serving as its judge in Shawwal of the year 462 AH (1070 CE). Yahya ibn Sa'id ibn al-Hadidi prayed over him. Ibn Mutahir mentioned part of this.”

The Book of the Categories of Nations

الْأُمَمُ طَبَقَاتِ كِتَابُ

الْأَنْدَلُسِيُّ صَاعِدِ بْنِ أَحْمَدَ بْنِ صَاعِدِ الْقَاسِمِ أَبِي الْقَاضِي

Chapter One: The Ancient Nations

الرَّحِيمَ الرَّحْمَنَ اللَّهُ بِسْمِ
يَسِّرْ رَبِّ

وَمَعَارِبَهَا الْأَرْضِ مَشَارِقِ فِي النَّاسِ جَمِيعَ أَنَّ أَعْلَمَ تَعَالَى: اللَّهُ رَحِمَهُ صَاعِدِ بْنِ أَحْمَدَ بْنِ صَاعِدِ الْقَاسِمِ أَبُو الْقَاضِي قَالَ
وَاللُّغَاتِ. وَالصُّوَرِ بِالْأَخْلَاقِ أَشْيَاءَ: بِثَلَاثَةِ يَتَمَيَّزُونَ وَاحِدًا نَوْعًا كَانُوا وَإِنْ وَشَمَالَهَا وَجَنُوبَهَا¹

²الْقَدِيمَةُ [الْأُمَمُ : الْأَوَّلُ] [البَابُ]

سَالَفٍ فِي كَانُوا النَّاسِ أَنَّ الْقُرُونِ طَبَقَاتٍ عَنْ وَفَحَصَ الْأَجْيَالِ سَائِرٍ عَنْ وَبَحَثَ الْأُمَمِ بِأَخْبَارٍ عَنِّي مَنْ وَزَعَمَ
وَحَدُّ الْمَعْمُورِ الْوَسْطِ فِي مَسْكَنَهَا وَكَانَ الْفُرْسِ الْأُولَى (الْأُمَّةُ أُمَمٍ سَبْعَ اللُّغَاتِ وَافْتِرَاقِ الْقَبَائِلِ تَشَعَّبَ وَقَبْلَ الدَّهْورِ
وَقَاشَانَ وَقَمَّ وَهَمْدَانَ وَالدُّسْتُورُ وَالْكُجُجُ الْأَنْجَهَاتُ فِيهِ وَالَّذِي حُلُوَانَ بِعَقَبَةِ الْعِرَاقِ شِمَالِ فِي الَّتِي الْجِبَالِ مِنْ بِلَادِهَا
وَالرِّيِّ وَالشَّابَرَانَ وَأَرْزَنَ وَالْبَيْلِقَانَ وَمَوْلَتَانَ وَطَبْرِسْتَانَ أَذْرَبِيْجَانَ بِحَرِّ الْمُتَّصِلِ وَالْبَابِ أَرْمِينِيَّةَ إِلَى الْبِلَادِ مِنْ وَغَيْرِهَا
وَبُجَارَا. وَبَلَخَ وَخَوَارِزْمَ وَهَرَاةَ وَسَرَخْسَ وَالْمَرْوِ كَنْيْشَابُورَ خِرَاسَانَ بِلَادٍ إِلَى وَجُرْجَانَ وَالطَّالِقَانَ

[Translation:]

In the name of God, the Merciful, the Compassionate.

O Lord, make it easy!

The judge Abu al-Qasim Sa'id ibn Ahmad ibn Sa'id—may God the Exalted have mercy on him—said: Know that all people in the East and West of the Earth, in its South and North, though they are one species, are distinguished by three things: by their moral character, their physical forms, and their languages.

Those who have concerned themselves with the reports of nations and investigated the various generations, and examined the categories of ages, say that in earliest times, before the branching of tribes and the divergence of languages, there were seven nations:

The First Nation: The Persians. Their dwelling was in the middle of the inhabited world. Their lands include the mountains in the north of Iraq, the region of Hulwan, and the area containing the

¹The author here sets forth the fundamental unity of humankind while acknowledging three sources of differentiation: moral character (*akhlāq*), physical appearance (*ṣuwar*), and language (*lughāt*). This tripartite division reflects the anthropological understanding of the 11th century.

²Chapter One: The Ancient Nations. These are the peoples who lived in the earliest ages of the world, before the rise of the great civilizations of the Near East.

Anjahat, Kuraj, Dustur, Hamadan, Qumm, Qashan, and other lands extending to Armenia, the Gate connecting to the Sea of Azerbaijan, Tabaristan, Multan, Bilaqan, Arzan, Shaboran, al-Rayy, Taliqan, and Jurjan, to the lands of Khurasan—Nayshabur, Marw, Sarakhs, Harat, Khwarizm, Balkh, and Bujara.

Chapter Two: Science in India

الهند في العلم
 علينا وردت ولا علومهم من طرف إلينا نصل فلم نألفهم فلم يبيننا المالك واعتراض بلادنا من الهند ولبعده
 علمهم. من بالقليل إلا سمعنا ولا مذهبهم من نبد إلا
 ومذهب الأزيج ومذهب هند السند مذهب وهو عنهم المشهورة الثلاثة المذهب النجوم علوم في الهند مذهب فن
 والقوام الإسلام من جماعة تقلده الذي المذهب وهو هند السند مذهب إلا التحصيل على منهم إلينا يصل ولم الأركند
 محمد بن والحسين الخوارزمي موسى بن ومحمد البغدادي الله عبد بن وحبش الفزاري إبراهيم بن كمحمد الأزيج فيه
 زيجيه. في الآدمي بن الحسين حكى كذلك الداهر >> >> الدهر هند السند وتفسير وغيرهم. الآدمي بابن المعروف

[فصل]

الكواكب أن ذكروها التي مذهبهم فإن العالم مدة عدد ألا هند السند أصحاب وافقوا فإنهم هند السند أصحاب وأما
 والجوزهرات والأوجات الكواكب حركات أبدأت الأمر ذلك فمئذ الحمل رأس في مجتمعت وجوزهراتها وأوجاتها
 فإذا البروج في والجوزهرات والأوجات الكواكب تتفرق حتى طويلاً دهرًا خراباً السفلى العالم وبقي الأرض في
 الكواكب من واحد ولكل عندهم. غاية غير إلى أبداً هكذا الأول الأمر إلى السفلى العالم حالة بدأت كذلك كان
 ذكرتها. قد العالم مدة عندهم هي التي المدة هذه في ما أدوار والجوزهرات والأوجات

[Translation: Science in India]

Because of the distance of India from our lands and the obstacles between us and them, we have not become familiar with them. No portion of their sciences has reached us, nothing has come to us but fragments of their doctrines, and we have heard only a little of their knowledge.

Among the doctrines of the Indians in the science of astrology, three doctrines are famous among them: the doctrine of Sindhind, the doctrine of the Azij, and the doctrine of Arkand. Only the doctrine of Sindhind has reached us in a complete form. It was followed by a group of Muslims, and those who established it were al-Azyaj such as Muhammad ibn Ibrahim al-Fazari, Habash ibn Abd Allah al-Baghdadi, Muhammad ibn Musa al-Khwarizmi, al-Husayn ibn Muhammad known as Ibn al-Adami, and others. The interpretation of Sindhind is “the Eternal Time” (al-Dahr al-Dahir), as al-Husayn ibn al-Adami related in his zij.

[Section]

As for the followers of Sindhind, they agree that the duration of the world has no fixed number. Their doctrine which they mention is that the planets, their apogees, and their nodes were all gathered at the head of Aries. Since that event, the movements of the planets, apogees, and nodes began. The lower world remained desolate for a long time until the planets, apogees, and nodes dispersed through the zodiac signs. When that happened, the condition of the lower world returned to its first state, and this continues without end according to them. Each of the planets, apogees, and nodes has cycles during this period, which they consider the duration of the world, which I have mentioned.

Chapter Three: Science among the Persians

الْفُرسُ فِي الْعِلْمِ
 مُلُوكًا وَأَنْوَسَهَا أَقْلِيمًا وَأَشْرَفَهَا دَارًا الْأُمَمِ وَأَوْسَطِ الشَّائِخِ وَالْعِزِّ الْبَادِخِ الشَّرَفِ فَاهْلُ الْفُرسِ وَهِيَ الثَّانِيَةُ الْأُمَّةُ وَأَمَّا
 غَارُهُمْ مَنْ بِهِمْ وَتَغَلَّبَ نَاوَاهُمْ مَنْ عَنْهُمْ تُحَايِي وَرُؤُوسُ تَجْمَعُهُمْ مُلُوكٌ لَهُمْ وَكَانَتْ الْمُلْكُ لَهَا دَامَ غَيْرُهُمْ أُمَّةٌ نَعْلَمُ وَلَا
 ذَلِكَ يَأْخُذُ التَّنْظِيمَ وَأَحْسَنَ وَدَوَامِ اتِّصَالٍ عَلَى حَظِّهِمْ فِيهِ مَا عَلَى الْأُمُورِ مِنْ وَتَحْتَمِلُهُمْ مَظْلُومِهِمْ عَنْ ظَالِمِهِمْ وَتَدْفَعُ
 سَالِفِهِمْ. عَنْ وَغَائِبِهِمْ أَوْلَهُمْ عَنْ آخِرِهِمْ

وَالْعَجَمِ [الْعَرَبِ] فِي
 الْفُرسِ مَنْوُشِهْرَ مُلْكٍ إِلَى نُوحِ بْنِ سَامِ بْنِ الْأَذِ بْنِ أَمِيمِ بْنِ كَيْنُمُورَثَ مُلْكٍ ابْتِدَاءً مِنْ أَنَّ ذَلِكَ فِي قِيلَ مَا وَأَصَحُّ
 الْفُرسِ مُلُوكٍ مِنَ الثَّانِيَةِ الطَّبَقَةِ مُلُوكٍ أَوَّلُ مَنْوُشِهْرَ مُلْكٍ ابْتِدَاءً إِلَى السَّلَامِ عَلَيْهِ الْبَشَرِ أَبُو آدَمَ عِنْدَهُمْ هُوَ الَّذِي كَانَهُمْ
 الْفُرسِ مُلُوكٍ مِنَ الثَّلَاثَةِ الطَّبَقَةِ مُلُوكٍ أَوَّلِ دُوعِ بْنِ كَيْثُبَادَ مُلْكٍ ابْتِدَاءً إِلَى مَنْوُشِهْرَ مُلْكٍ وَمِنْ كَامِلَةٍ. سَنَةِ أَلْفٍ نَحْوِ
 عِنْدَ ذَلِكَ الْفُرسِ مُلُوكٍ مِنَ الرَّابِعَةِ الطَّبَقَةِ وَهِيَ الطَّوَائِفُ مُلْكٍ ابْتِدَاءً إِلَى كَيْثُبَادَ مُلْكٍ وَمِنْ عَامٍ. مَايَتِي مِنْ قَرِيبٍ
 إِلَى الطَّوَائِفُ مُلْكٍ أَوَّلِ وَمِنْ سَنَةِ. أَلْفٍ نَحْوِ الْفُرسِ مُلُوكٍ مِنَ الثَّلَاثَةِ الطَّبَقَةِ مُلُوكٍ آخِرِ دَارًا بْنِ لِدَارًا الْإِسْكَندَرِ مُقْتَلٍ
 سَنَةِ خَمْسِمِائَةِ الْفُرسِ مُلُوكٍ مِنَ الْخَامِسَةِ الطَّبَقَةِ وَهِيَ إِسْفَانِيَارَ بْنِ مُلُوكٍ أَوَّلِ السَّاسَانِيَّ بَابَكِ بْنِ أَزْدَشِيرَ مُلْكٍ ابْتِدَاءً
 قَتْلَ عِنْدَ ذَلِكَ الْأَرْضِ مِنَ الْفُرسِ دَوْلَةَ انْقِضَاءً إِلَى بَابَكِ بْنِ أَزْدَشِيرَ مُلْكٍ ابْتِدَاءً وَمِنْ سَنَةِ. وَسِتُونَ وَإِحْدَى

[Translation: Science among the Persians]

As for the second nation, the Persians: they are a people of towering nobility and exalted glory, the most settled of nations and the noblest in region, and the most prosperous in kings. We know of no nation besides them whose sovereignty endured so long, who had kings that united them and leaders that defended them from those who opposed them, overcame with them those who envied them, and protected their oppressed from their oppressors, and who endured affairs in a way that secured their portion with continuity and permanence and the finest organization, so that the last of them took this over from the first, and the absent from the absent.

[On the Arabs and the Non-Arabs]

The most correct opinion on this is that from the beginning of the reign of Kayumars ibn Amim ibn Aladh ibn Sam ibn Nuh (Noah) to the reign of Manushihr the Persian—who is, according to them, Adam the father of mankind, peace be upon him—to the beginning of the reign of Manushihr, the first king of the second class of Persian kings, was about a thousand complete years. From the reign of Manushihr to the beginning of the reign of Kaythubadh ibn Dua, the first king of the third class of Persian kings, was about two hundred years. From the reign of Kaythubadh to the beginning of the reign of the Tribes (al-Tawa'if), the fourth class of Persian kings, at the killing of Alexander (al-Iskandar) of Dara ibn Dara, the last king of the third class of Persian kings, was about a thousand years. From the beginning of the reign of Azdashir ibn Babak the Sasanian, the first king of the Banu Isfaniyar (the fifth class of Persian kings), was five hundred and sixty-one years. From the beginning of the reign of Azdashir ibn Babak to the end of the Persian state on earth, at the killing [of Yazdagird]...

Chapter Four: Science in Greece

اليونان في العلم
 شَطَّ مِنْ مَمْدَّةٍ مُلْكِهِمْ دَارَةٌ وَكَانَتْ بِهَا. الْمُحِيطَةُ وَالْأَمَمُ وَقِسْطَنْطِينِيَّةَ وَالرُّومَ فَلَسْطِينَ أَهْلُ فَهُمْ الْيُونَانِيُّونَ وَأَمَّا
 مَدِينَةُ الْعُظْمَى مَدِينَتُهُمْ وَكَانَتْ الشَّامَ. بَحْرًا إِلَى الْقُطْبِ أَقْصَى وَمِنْ الْمَغَارِبِ، فِي شَطِّهِ إِلَى الْمَشَارِقِ فِي الْمَتَوَسِّطِ بَحْرٍ
 مُتَنَوِّعَةٍ. كَثِيرَةٌ وَعُلُومُهُمْ جَدًّا قَدِيمَةٌ دَوْلَتُهُمْ وَكَانَتْ الْإِسْكَندَرِيَّةُ.
 وَبَرْقَلِسُ وَهَبَاطِيَا وَجَالِينُوسُ وَبَطْلِيمُوسُ وَارِسْطُو وَأَفْلَاطُونُ فَالَيْسُ عِنْدَهُمْ وَالْحِكْمُ الْعُلُومُ فِي تَكَلُّمٍ مِنْ أَوَّلٍ وَكَانَ
 بِإِصْلَاحٍ قَامُوا الدِّينَ وَهُمْ وَالْحِكْمُ. الْعُلُومُ أَهْلٍ مِنْ هَؤُلَاءِ إِلَى يُنْسَبُ مِنْ وَجْمِعٍ وَأَرْشِمِيدُسُ وَنُقُومَاخُسُ وَأَقْلِيدُسُ
 الْعَرَبِ لُغَةً إِلَى نَقَلَتْ إِنَّمَا ثُمَّ بَلَّغَتْهُمْ. ذَلِكَ فِي مُؤَلَّفَاتِهِمْ وَكَانَتْ قَوَاعِدُهَا وَتَقْيِيدُ أَقْسَامِهَا وَتَرْتِيبُ وَتَوْضِيحُهَا الْعُلُومُ أُصُولُ
 الشَّرْقِ. فِي النَّصَارَى مِنَ الدِّينِ أَهْلُ وَاتَّخَذَهَا الْمَنَازِرَةَ نَشَرَهَا أَنْ بَعْدَ

[Translation: Science in Greece]

As for the Greeks, they are the people of Palestine, Rome, Constantinople, and the surrounding nations. Their kingdom extended from the shore of the Mediterranean Sea in the East to its shore in the West, and from the far north to the Syrian Sea. Their greatest city was the city of Alexandria. Their state was very ancient and their sciences were many and varied.

The first to discourse on the sciences and wisdom among them were Thales, Plato, Aristotle, Ptolemy, Galen, Hippocrates, Proclus, Euclid, Nicomachus, Archimedes, and all those attributed to these in the sciences and wisdom. They were the ones who established the principles of the sciences, elucidated them, arranged their parts, and codified their rules, and their compositions on this were in their own language. Then they were translated into Arabic after the Manadhirah disseminated them and the religious scholars among the Christians in the East adopted them.

Chapter Five: Science in Egypt

مِصْرَ فِي الْعِلْمِ
عَهْدٍ فِي مَنَّهُمُ الْيُونَانِيُّونَ أَخَذَتْهَا مِنْذُ تَبْلُغَ لَمْ وَلَكِنَّهَا نَافِعَةٌ قَدِيمَةٌ عُلُومُهُمْ وَكَانَتْ الْعِلْمُ مِصْرَ أَهْلُ فَهْمُ الْقِبْطُ وَأَمَّا
عَنْ فَأَخَذُوا الْأُمَمَ جَمِيعَ مِنَ الْعُلَمَاءِ إِلَيْهَا وَحَشَدَ لِلْعُلُومِ مَوْطِنًا وَجَعَلَهَا الْإِسْكَندَرِيَّةَ مَدِينَةً بَنَى الَّذِي الْمَلِكِ بَطْلِيمُوسَ
تَأْلِيفَ وَعِلْمُ الْهَيْئَةِ عِلْمُ ذَلِكَ وَمِنْ فِيهَا. وَصَنَفُوا وَنَقَّحُوا إِلَيْهَا أَضَافَ مَا الْيُونَانِ حَكَمَ مِنْ فِيهَا فَأَدْخَلُوا عُلُومَهُمُ الْقِبْطُ
الْعُلُومِ. مِنْ ذَلِكَ وَغَيْرِ وَالْأَدْوِيَّةِ الْعِلَلِ وَعِلْمُ الْكَوَاكِبِ

[Translation: Science in Egypt]

As for the Copts, they are the people of Upper Egypt. Their sciences were ancient and beneficial, but they did not advance further since the Greeks took them from them in the time of Ptolemy the king who built the city of Alexandria and made it a center for the sciences, gathering scholars from all nations. They took from the Copts their sciences and incorporated into them Greek wisdom, adding to them, refining them, and composing works on them. Among these were the science of astronomy, the science of the configuration of the planets, the science of causes and remedies, and other sciences.

Chapter Six: Science among the Jews

الْيَهُودِ فِي الْعِلْمِ
 بِلَادِهِمْ مِنْ خَرَجُوا لَمَّا إِنَّهُمْ ثُمَّ وَمُلُوكِهِمْ. أَنْبِيَائِهِمْ زَمَنٍ فِي الْأُولَى أَرْزَمْتِهِمْ فِي عُلُومٍ لَهُمْ كَانَتْ فَقَدْ الْيَهُودُ وَأَمَّا
 قَدْ وَلَكِنَهُمُ الْعِلْمُ. عَلَى مُوَظَبَةٍ وَلَا بَقَاءِ أَهْلٍ يَكُونُوا لَمْ لِأَنَّهُمْ وَتَنَكَّرَتْ عُلُومُهُمْ ضَاعَتْ وَالْعَرَبُ الْعَجَمُ بِلَادٍ فِي وَتَفَرَّقُوا
 الْأَنْدَلُسِيِّ مَيْمُونِ بْنِ مُوسَى عُلَمَائِهِمْ أَشْهَرُ مِنْ وَكَانَ وَالتَّوْحِيدِ. وَالشَّرَائِعِ التَّعَبُّدِ فِي عُلُومِ الْأَنْبِيَاءِ تَوَارِيثٍ مِنْ حَفِظُوا
 وَغَيْرَهَا. وَالطَّبِّ وَالْفَلَسَفَةِ الْكَلَامِ عِلْمٍ فِي صَنَفِ الَّذِي

[Translation: Science among the Jews]

As for the Jews, they had sciences in their earliest times, in the era of their prophets and kings. But when they left their lands and dispersed into the lands of the non-Arabs and the Arabs, their sciences were lost and forgotten because they were not a people of endurance or perseverance in science. However, they did preserve from the inheritance of the prophets sciences concerning worship, religious laws, and monotheism. Among their most famous scholars was Musa ibn Maymun (Maimonides) al-Andalusi, who composed works on the science of theology, philosophy, medicine, and other subjects.

Chapter Seven: Science among the Arabs

الْعَرَبُ فِي الْعِلْمِ
 مَعِيشَتِهِمْ لَصِعْرٍ عَلَيْهِ تَوَاضَعُوا وَلَا بِالْعِلْمِ تَعْنَى لَا مِنْهُمْ الْبَادِيَةُ وَكَانَتْ حَاضِرَةً. وَأَمَّةٌ بَادِيَةٌ أَمَّةٌ أَمْتَانِ: فَهُمْ الْعَرَبُ وَأَمَّا
 وَغَيْرُ النَّحْوِ وَالشَّعْرِ وَالسَّحْرِ وَالنُّجُومِ الطَّبِّ فِي الْجَاهِلِيَّةِ فِي عُلُومِهِمْ كَانَتْ فَقَدْ الْحَاضِرَةُ وَأَمَّا الْبِلَادُ. فِي وَتَقْلِهِمْ
 الْعُلُومِ طَلَبِ فِي الْعَرَبِ اجْتَهَدَ وَالْفَلَاحِ لِلنَّجَاةِ سَبِيًّا وَجَعَلَهُ عَلَيْهِ وَحَثَّ الْعِلْمِ طَلَبِ فِي وَشَرَعَ الْإِسْلَامُ جَاءَ لَمَّا ثُمَّ ذَلِكَ.
 وَفَنُونَ. عُلُومِ أَهْلٍ قَلِيلٍ زَمَنٍ فِي أَصْبَحُوا حَتَّى الْعَرَبِ بُلْغَةً وَتَصْيِيرَهَا الْأُخْرَى الْأُمَمِ مِنْ وَنَقْلَهَا
 فِيهَا وَصَنَفُوا فَتَقَلُّوْهَا الْأَوَائِلِ لِعُلُومِ تَعَرَّضُوا إِنَّهُمْ ثُمَّ وَالتَّفْسِيرِ وَالْحَدِيثِ وَالْفَقْهِ الْكَلَامِ عِلْمٌ إِلَيْهِ نَشَطُوا مَا أَوَّلُ وَكَانَ
 الطَّبِّ وَفِي الْفَلَكَيِّ. مَعَشَرٍ وَأَبُو وَالبَّتَانِيُّ الْفَرَّغَانِيُّ النُّجُومِ عِلْمٌ فِي الْعَرَبِ أَشْهَرُ مِنْ وَكَانَ كَثِيرَةً. أَشْيَاءٌ إِلَيْهَا وَأَضَافُوا
 الْعِلْمِ وَفِي الطُّوسِيِّ. الدِّينِ وَشَرَفُ وَالبَّتْرُوجِيُّ الْمَجْرُطِيُّ بِطَلِيمُوسُ الْهَيْئَةِ وَفِي رِشْدٍ. وَابْنُ وَالمَجُوسِيِّ سِينَا وَابْنُ الرَّازِيِّ
 رِشْدٍ. وَابْنُ سِينَا وَابْنُ وَالفَارَابِيُّ الْكِنْدِيُّ الْفَلَسَفَةِ وَفِي وَالكِشِّيُّ. الْخَوَارِزْمِيُّ الْحَسْبِ

[Translation: Science among the Arabs]

As for the Arabs, they are two nations: a desert nation and a settled nation. The desert among them did not concern itself with science nor persevere in it due to the hardship of their livelihood and their movement from land to land. As for the settled people, they had sciences in the Jahili period in medicine, astrology, magic, poetry, grammar, and other subjects. Then when Islam came and legislated the pursuit of knowledge, encouraged it, and made it a means of salvation and success, the Arabs strove in seeking the sciences, translating them from other nations, and rendering them into Arabic, until in a short time they became a people of sciences and arts.

The first [sciences] they were active in were theology (*kalam*), jurisprudence (*fiqh*), *hadith*, and [Qur'anic] exegesis (*tafsir*). Then they turned to the sciences of the ancients, translating them, composing works on them, and adding much to them. Among the most famous Arabs in the science of astrology were al-Farghani, al-Battani, and Abu Ma'shar the astronomer. In medicine: al-Razi, Ibn Sina (Avicenna), al-Majusi, and Ibn Rushd (Averroes). In astronomy: Ptolemy al-Majriti, al-Bitruji (Alpetragius), and Sharaf al-Din al-Tusi. In arithmetic: al-Khwarizmi and al-Kashshi. In philosophy: al-Kindi, al-Farabi, Ibn Sina, and Ibn Rushd.

Editorial and Scholarly Notes

On the Manuscript and Its Preservation

The 1912 edition by Pere Louis Cheikho, S.J. (1859–1927) represents the first modern critical edition of this important work. Cheikho based his edition on a single manuscript preserved in the Oriental collections of Europe. The manuscript, as Cheikho describes it, is written in a clear hand resembling Persian script on thick yellowed paper, bound skillfully in gold-colored paper.

The *Kitab Tabaqat al-Umam* has survived in a relatively small number of manuscripts compared to other major works of the Islamic scientific tradition. This may be attributed to the fact that it was composed in al-Andalus and did not circulate as widely in the eastern Islamic lands as works produced in Baghdad or other eastern centers.

On the Author

Sa'id al-Andalusi (1029–1070 CE), whose full name is Abu al-Qasim Sa'id ibn Ahmad ibn Sa'id al-Taghlabi, was a jurist and scholar of Toledo. Born in Almeria (al-Mariyya) in 420 AH, he served as a judge (*qadi*) in Toledo under the Hammudid amir Yahya ibn Dhi al-Nun. He died in Shawwal 462 AH (July 1070 CE).

On the Structure of the Work

The *Kitab Tabaqat al-Umam* is divided into two main parts:

1. **The Ancient Nations** (*al-Umam al-Qadima*): Seven nations that existed in earliest times—the Persians, the Chaldeans, the Greeks, the Egyptians, the Jews, the Arabs, and the Indians.
2. **The Islamic Nations** (*al-Umam al-Islamiyya*): A classification of the nations that embraced Islam, with special attention to their contributions to the sciences.

The work is notable for being one of the earliest attempts at a systematic history of science organized by civilizations, predating similar European efforts by several centuries.

On the Scientific Content

Al-Andalusi's work contains valuable information about:

- The state of astronomy and astrology in various ancient civilizations
- The transmission of scientific knowledge between cultures
- The role of translation in preserving Greek scientific heritage
- Early Indian contributions to mathematics and astronomy (including the decimal system)
- The development of scientific institutions such as the House of Wisdom in Baghdad

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REFERENCE TEXT

al-Muqaddasi

Ahsan al-Ta'asim fi Ma'rifat al-Aqalim

Selections from M. J. de Goeje's 1906 edition

Arabic text with English rendering and editorial apparatus

This reader presents selected Arabic reference passages from the 1906 edition with English renderings and basic editorial context.

Author's Preface

الرَّحِيمِ الرَّحْمَنِ اللَّهُ بِسْمِ

رَبِّ يَسِّرْ وَلَا تُعَسِّرْ بِفَضْلِكَ يَا كَرِيمُ

الْحَمْدُ لِلَّهِ الَّذِي خَلَقَ الْفَلَكَ وَصَوَّرَ فَاتَقَنَّ صُنْعَ الْمَرِيَّةِ بِلَا مَشِيرٍ يُشَاوِرُهُ، وَدَبَّرَهَا بِلَا مُعِينٍ يُعَايِدُهُ، أَتَقَنَّا أَتَقَنَّا، وَأَحْكَمَهَا بِلَا إِعْوَانٍ، أَوْتَدَّ الْأَرْضَ بِالرَّاسِيَّاتِ لَثَلَا تَمِيدَ، وَأَحَاطَهَا بِالْبَحْرِ كَيْلًا يَغْلِبُ مَاؤُهَا وَيُرِيدُ، وَبَثَّ فِيهَا عِبَادَهُ لِيَنْظُرَ كَيْفَ يَعْمَلُونَ فَمِنْهُمْ مَنْ أَحْسَنَ وَافْتَدَى، وَمِنْهُمْ مَنْ كَفَرَ وَلَسِيَ وَصَلَّى اللَّهُ عَلَى خَاتِرِ الْأَنْبِيَاءِ، وَأَكْرَمِ الْأَوْلِيَاءِ، مُحَمَّدٍ وَعَلَى آلِهِ وَصَحْبِهِ وَسَلَّمْ تَسْلِيمًا كَثِيرًا.

In the name of God, the Compassionate, the Merciful.

My Lord, make it easy and do not make it difficult, by Thy grace, O Generous One.

Praise be to God, who created the celestial orb and formed it with consummate skill, without any counsellor to consult; who ordered it without any assistant to help; who perfected it in the most perfect manner, and established it without need of support; who fixed the earth firmly lest it should sway, and surrounded it with the sea as a barrier protecting its waters; and who dispersed His servants upon it that He might observe how they act—and among them are those who do good and are redeemed, and among them are those who disbelieve and forget. And may God bless the Seal of the Prophets, the noblest of the chosen, Muhammad, and his family and companions, with abundant peace.

قَالَ أَبُو عَبْدِ اللَّهِ مُحَمَّدُ بْنُ أَحْمَدَ الْمُقَدَّسِيُّ: أَمَّا بَعْدُ، فَإِنَّهُ مَا زَالَتِ الْعُلَمَاءُ تَرْغَبُ فِي تَصْنِيفِ الْكُتُبِ لِمَا لِحَرَسِهِمْ، وَلَا تَقْطَعُ أَخْبَارُهُمْ، فَأَحْبَبْتُ أَنْ أَتَّبِعَ سُنَّتَهُمْ، وَأَقْفُو سُنَّتَهُمْ، وَأَقِيمَ عِلْمًا أُحْيِي بِهِ ذِكْرِي، وَأَشِيرَ بِهِ وَفِيلَةَ، وَوُجِدَتِ الْعُلَمَاءُ قَدْ سَبَقُوا إِلَى الْعُلُومِ فَتَمَلَّقُوا عَلَى الْإِجْتِهَادِ، ثُمَّ تَبِعْتَهُمُ الْأَخْلَافُ فَشَرَحُوا كَلَامَهُمْ وَاخْتَصَرُوهُ، فَرَأَيْتُ أَنْ أُولَفَ كِتَابًا فِي أَخْبَارِ الْأَقَالِيمِ، وَأَشْكَالِ الْأَرْضِ، وَأَنْوَاعِ الْعِبَادَاتِ، وَمَسَالِكِ الْمَمَالِكِ، وَالْبِلَادِ، وَأَمْصَارِهَا الْمَشْهُورَةِ، وَمَدَائِنِهَا الْمَعْرُوفَةِ، وَمَنَازِلِهَا الْمَسْلُوكَةِ، وَطُرُقِهَا الْمُسْتَعْمَلَةِ، وَعَنَاصِرِ الْعَشَائِرِ وَالْأَلَاتِ، وَمَعَادِنِ الْجَمَلِ وَالتِّجَارَاتِ، وَاخْتِلَافِ أَهْلِ الْبُلْدَانِ فِي كَلَامِهِمْ وَأَصْوَاتِهِمْ وَأَسْنَانِهِمْ وَالْوَانِيهِمْ، وَمَذَاهِبِهِمْ وَمَكَاتِبِهِمْ وَأَوْزَانِهِمْ، وَنُقُودِهِمْ، وَأَفْضِيَةِ الْمَعَاشِ لِأَهْلِ كُلِّ بَلَدٍ، وَأَشْيَاخِهِمْ، وَعِلْبَائِهِمْ، وَزُهَادِهِمْ، وَحُكَّامِهِمْ.

Abu Abdallah Muhammad ibn Ahmad al-Muqaddasi said: Now then, the learned have always been eager to compose books that fill the scope of their investigations, and their reports are inexhaustible. I desired to follow their example and to walk in their footsteps, and to raise a standard whereby my memory might be kept alive and my name remembered. I found that the learned had already preceded me in the sciences and had distinguished themselves through diligent effort, and that successors had followed them, explaining and abridging their works. So I resolved to compose a book on the reports of the climes, the forms of the earth, the varieties of religious practices, the routes of kingdoms and lands, their famous garrison towns and well-known cities, their frequented

stations and well-trodden roads; the elements of tribes and families, the mines of gems and trade goods; the differences among the peoples of the lands in their speech, voices, teeth, and colors; their doctrines, scripts, and weights; their coins and the means of livelihood for the people of each land; their elders, scholars, ascetics, and sages.

Necessary Preliminaries

مُقَدِّمَاتٌ وَفُصُولٌ لَا بَدْ مِنْهَا

أَعْلَمُ أَنِّي أَسَّسْتُ هَذَا الْكِتَابَ عَلَى قَوَاعِدَ مُحْكَمَةٍ وَأَسْنَادَاتٍ بِدَعَائِمٍ قَوِيَّةٍ، وَتَخَرَّجْتُ فِيهِ جِهَادَ الصَّوَابِ، وَاسْتَعَنْتُ فِيهِمْ أُولِي الْأَلْبَابِ، وَاسْمِهِ أَنْ يُجْتَنَبَ فِيهِ الْخَطَأُ وَالزَّلَلُ، وَيَبْلُغَ فِيهِ الرَّجَاءُ وَالْأَمَلُ، فَعَلَى قَوَاعِدِهِ أَوْصَفُ بِفَنِّيَّاتِهِ مَا شَاهَدْتُهُ وَعَقَلْتُهُ، وَحَرَفْتُهُ وَعَلَّمْتُهُ، وَعَلَيْهِ رَفَعْتُ الْبُنْيَانَ وَالْأَرْكَانَ، وَمِنْ قَوَاعِدِهِ أَيْضًا وَإِرْكَانِهِ، وَمَا اسْتَعَنْتُ بِهِ عَلَى تَبْيِينِهِ، سَوَالُ ذَوِي الْعُقُولِ مِنَ النَّاسِ، وَمَنْ لَمْ أَعْرِفْهُمْ بِالْغُلَّةِ وَالْإِتْبَاسِ، عَنِ الْكَبِيرِ وَالْأَعْمَالِ فِي الْأَطْرَافِ الَّتِي بَعُدَتْ عَنْهَا، وَلَمْ يَتَقَدَّرْ لِي الْوُصُولُ إِلَيْهَا، فَمَا وَقَعَ عَلَيْهِ اتِّفَاقُهُمْ أَثْبَتُهُ، وَمَا اخْتَلَفُوا فِيهِ نَبَذْتُهُ، وَمَا لَمْ يَكُنْ لِي بَدَلٌ فِي اسْتِنَادَاتِهِ إِلَى الْمَاضِي ذَكَرْتُ أَوْ قُلْتُ زَعَمُوا، وَشَاهَدْتُهُ بِفُصُولٍ وَجَدْتُهُ فِي خَزَائِنِ الْمُلُوكِ.

وَكُلُّ مَنْ سَبَقَنَا إِلَى هَذَا الْعِلْمِ لَمْ يَسْلُكِ الطَّرِيقَ الَّتِي قَصَدْنَاهَا، وَلَا طَلَبَ الْغَوَائِدَ الَّتِي أَرَدْنَاهَا، أَمَّا أَبُو عَبْدِ اللَّهِ الْبَكْرِيُّ فَكَانَ دَزِيرَ أَمِيرِ خُرَاسَانَ، وَكَانَ صَاحِبُ فَلَاسَفَةٍ وَنَحْوِهِمْ وَهَيْئَةً مُجْتَمِعِ السُّفَرَاءِ وَسَالَهُمْ عَنِ الْمُلُوكِ وَدَخَلَهَا.

Know that I have founded this book upon sound principles and solid bases, and I have exerted myself therein to attain correctness, seeking the help of those of understanding. I strove to avoid error and mistake, and to achieve the hoped-for goal. On these principles I describe in its chapters what I have witnessed and understood, recorded and learned; upon them I raised the structure and pillars. Among its principles and pillars, and what I relied upon to elucidate it, was to question men of intellect, and those whom I did not know out of negligence and obscurity, about the great matters and the works in the distant regions which I could not reach. Whatever they agreed upon I confirmed; whatever they differed upon I rejected; and what I had no way to verify by reference to the past, I recorded with "they say" or "they claim." And I witnessed it in detail and found it in the treasures of kings.

Everyone who preceded us in this science did not follow the path we intended, nor seek the principles we desired. As for Abu Abdallah al-Bakri, he was the vizier of the governor of Khurasan, a philosopher, and he gathered the ambassadors and questioned them about the kings, and entered their lands.

Chapter on the Seas and Rivers

ذِكْرُ الْبَحَارِ وَالْأَنْهَارِ

أَعْلَمُ أَنَا لَمْ نَرِ فِي الْإِسْلَامِ إِلَّا بَحْرَيْنِ حَسَبُ أَحَدِهِمَا يَخْرُجُ مِنْ نَحْوِ مَشَارِقِ الشَّامِ بَيْنَ بِلَادِ الصِّينِ وَبِلَادِ السَّوَادِ، فَإِذَا بَلَغَ مَمْلَكَةَ الْإِسْلَامِ دَارَ عَلَى جَزِيرَةِ الْعَرَبِ، كَمَا مَثَلْنَاهُ، وَلَهُ خَلِيجَانِ كَثِيرَةٌ وَشَعْبٌ عَدَّةٌ، وَقَدْ اخْتَلَفَ النَّاسُ فِي وَصْفِهِ وَالْمَصِيرِ إِلَى فَنَائِهِمْ مِنْ جَاعِلَةٍ، شَبِيهَةٌ طَبَسِيَّةٍ بِخَلِيجِ بَنِيَانٍ يَطُوفُ عَلَى الْقَسَمِ وَطَرَفِ بِالْعِرَاقِ وَطَرَفِ بَعْمَانَ، وَخَلِيجِ الشَّيْنِ وَبِلَادِ الشَّيْنِ حَدِيثَةٌ، وَذَنْبُهُ بِالْعِرَاقِ، وَعِنْدَهُ خُرَاسَانَةُ أَمِيرِ خُرَاسَانَ وَرَأْيُهُ مُتَّفَقٌ عَلَى وَرَقَةٍ فِي خَزَانَةِ أَمِيرِ الْمُؤْمِنِينَ بَنِيْسَابُورَ وَفِي خُرَاسَانَةِ عِنْدَ الدَّوْلَةِ وَالصَّاحِبِ، وَإِذَا كُلُّ مِثْلٍ، فَإِنَّا لَمْ نَعْرِفْهَا، وَمَا أَنَا فَسَّرْتُهُ بِمُخْتَلَفِ الْأَثَارِ وَإِذَا فِي بَعْضِهِنَّ خَلِيجَانِ يَعْتَمِدُ عَلَى الْقَلِيلِ مِنْ كُلِّهَا، وَأَمَّا النَّهْرُ الْأَعْظَمُ فَجَرَاهُ فِي الْيَابَسَةِ.

Know that in the Islamic lands we have not seen [bodies of water] other than two seas, according to one of which: it emerges from the direction of the eastern regions of Syria, between the lands of China and the lands of the Sawad. When it reaches the realm of Islam, it encircles the Arabian Peninsula, as we have represented it. It has many gulfs and numerous branches. The people have differed in their descriptions of it and in [tracing] the extent of its reaches: some hold it to be like the Tabi Tabasiyyah in the Bay of Baniyan, which runs along the division, with one end in Iraq and one end in Oman; the Bay of China and the lands of China are near it, and its tail extends toward Iraq, and nearby is Khurasan of the governor of Khurasan. His opinion is confirmed by a map in the treasury of the Commander of the Faithful in Nishapur, and in the Khurasan of the State and the Sahib. And so with every similar case—for we do not know them fully, and we have only explained them according to the varying accounts. Among them are bays that rely upon the calculation of longitude from all of them. As for the greatest river, its course lies in dry land.

Editorial Note on the Manuscripts

B = *codex Berolinensis* (Berlin manuscript)

C = *codex Constantinopolitanus* (Constantinople manuscript)

The editor, **M. J. de Goeje**, has based the present edition upon two principal manuscripts:

- a) **Codex Berolinensis** (B): A manuscript preserved in the Royal Library of Berlin, containing the complete text of al-Muqaddasi's work with certain lacunae.
- b) **Codex Constantinopolitanus** (C): A manuscript from Constantinople, fuller in certain respects, containing valuable additions and variants.

The apparatus criticus employs the following sigla:

Signum	Meaning
<i>B</i>	Codex Berolinensis
<i>C</i>	Codex Constantinopolitanus
<i>om.</i>	omits
<i>add.</i>	adds
<i>supra lin.</i>	above the line
<i>in marg.</i>	in the margin
<i>sine punctis</i>	without diacritical points
<i>pro</i>	reads (in place of)
<i>vid. infra</i>	see below
<i>vid. supra</i>	see above
<i>cod.</i>	codex (manuscript)

1 The Division of the Earth into Climes

وَقَدْ ذَكَّرْنَا أَنَّ مُلْكَ الْإِسْلَامِ حَسَبَ وَلَمْ تَتَكَلَّفْ مَسَالِكَ الْكُفَّارِ لِأَنَّهَا لَمْ تَكُنْ مِنْ بَابِنَا، وَقَدْ قَسَمْنَا أَرْضَ الْإِسْلَامِ إِلَى أَرْبَعَةِ عَشَرَ إِقْلِيمًا، وَأَفْرَدْنَا كُلَّ إِقْلِيمٍ فَصْلَنَاهُ وَأَعْنَقْنَاهُ، وَوَزَنَّا مَا فِيهِ مِنَ الْمَمَالِكِ، وَذَكَّرْنَا مَا فِيهِ مِنَ الْعَجَائِبِ، وَكَمْ أَقْلِيمُ الْعَرَبِ أَوْلَاهَا الْمَشْرِقُ الْحِجَازُ الْعِرَاقُ الشَّامُ مِصْرُ الْمَغْرِبُ.

We have mentioned that the realm of Islam is [divided] accordingly, and we have not concerned ourselves with the lands of the unbelievers, for they do not belong to our subject. We have divided the land of Islam into fourteen climes, and we have given each clime a separate chapter, and distinguished what it contains of kingdoms, and mentioned the wonders therein. The first clime is Arabia: the Hijaz, Iraq, Syria, Egypt, and the Maghrib.

1.1 The First Clime: Arabia

ذِكْرُ الْحِجَازِ:

وَالْحِجَازُ مِنْ أَفْضَلِ بِلَادِ الْعَرَبِ وَأَشْرَفِهَا، وَهُوَ مَحَلُّ أَمْصَارِ الْمُسْلِمِينَ وَمَوْضِعُ مَنَازِلِ الْعَرَبِ، وَمَوْلِدُ النَّبِيِّ صَلَّى اللَّهُ عَلَيْهِ وَسَلَّمَ وَمَنْبِتُ الشَّرَائِعِ، وَفِيهِ مَكَّةُ الْمُشْرِفَةُ وَالْمَدِينَةُ الْمُنُورَةُ وَالْيَمَنُ وَالْيَمَامَةُ، وَهُوَ أَقْلِيمٌ حَارٌّ قَلِيلُ الْمَاءِ وَالْعُشْبِ، وَأَكْثَرُهُ حِجَارَةٌ وَرِمَالٌ، وَالنَّاسُ فِيهِ بَدَوٌ وَحَضَرٌ، وَمَالُهُمُ الْإِبِلُ وَالْغَنَمُ، وَلَهُمْ مِنَ الْفَخْرِ وَالشَّجَاعَةِ وَالْحِمِيَّةِ مَا لَيْسَ لِغَيْرِهِمْ.

The Hijaz is the noblest and most honorable of the lands of Arabia; it is the dwelling-place of the Muslims and the abode of the Arabs, the birthplace of the Prophet (may God bless him and grant him peace) and the source of the revealed laws. In it are Mecca the Honored and Medina the Illumined, and Yemen and al-Yamama. It is a hot clime, with little water and herbage; most of it is stones and sand. The people are both Bedouin and settled; their wealth consists in camels and sheep; and they possess pride, courage, and valor such as no others have.

1.1.1 Mecca the Noble

مَكَّةُ الْمُكَرَّمَةُ:

وَمَكَّةُ أَشْرَفُ بِلَادِ اللَّهِ وَأَفْضَلُهَا، وَهِيَ مَوْضِعُ الْكَعْبَةِ الْمُعَظَّمَةِ وَالْمَسْجِدِ الْحَرَامِ، وَبِهَا أَبَدٌ فِي سِيرِ الْبِلَادِ، وَأَصِفُ مَا فِيهَا مِنَ الْعِمَارَاتِ وَالْمَسَاجِدِ وَالْأَسْوَاقِ وَالْأَحْيَاءِ وَالطَّرِيقَاتِ. وَمَكَّةُ فِي جَبَلٍ شَرِيفٍ بَيْنَ جِبَالٍ عَالِيَةٍ، وَهِيَ فِي سَفْحِ جَبَلٍ يُقَالُ لَهُ أَبُو قُبَيْسٍ، وَنَهْرُهَا صَعْبُ الْمَجْرَى، وَمَاؤُهَا مِنَ الْآبَارِ وَالْعَيُونِ، وَأَشْهَرُ مَا فِيهَا عَيْنُ زُبَيْدَةَ، وَهِيَ عَيْنٌ كَثِيرَةُ الْمَاءِ عَذْبَةُ الْمَذَاقِ.

Mecca is the noblest and most excellent of God's lands; it is the site of the venerated Kaaba and the Sacred Mosque. I begin my account of the cities with it, describing its buildings, mosques, markets, quarters, and roads. Mecca lies in a noble mountain among high peaks, on the slope of a mountain called Abu Qubays. Its river has a difficult course; its water comes from wells and springs, the most famous of which is the Spring of Zubayda—a spring abundant in water, sweet in taste.

Variant apparatus for this passage is omitted in this reader.

2 The Climes of Iraq and Syria

2.1 Iraq

ذِكْرُ الْعِرَاقِ:

وَالْعِرَاقُ هُوَ بِلَادُ كُورَةِ الْجَنَّةِ، وَمَوْضِعُ الْخِلَافَةِ وَالْمُلْكِ، وَبِهِ الْكُوفَةُ وَالْبَصْرَةُ وَوَاسِطُ الْأَنْبَارِ وَالرَّحْبَةُ وَحِصْنُ الْأَخْيَالِ، وَهُوَ أَرْضٌ خَصْبَةٌ كَثِيرَةُ الْمَاءِ وَالنَّخْلِ وَالْحِنْطَةِ وَالشَّعِيرِ، وَأَنْهَارُهُ كَثِيرَةٌ أَشْهَرُهَا دِجْلَةُ وَالْفُرَاتُ، وَالنَّاسُ فِيهِ أَكْثَرُهُمْ أَهْلُ بَصْرَةٍ وَعِلْمٍ وَفِقْهِ وَكَلَامٍ وَمَذَاهِبٍ.

Iraq is the land of the Courtyard of Paradise, the seat of the Caliphate and kingship. In it are Kufa, Basra, Wasit, Anbar, al-Rahba, and Hisn al-Ahyal. It is a fertile land, abundant in water, palm trees, wheat, and barley. Its rivers are numerous, the most famous being the Tigris and the Euphrates. Its people are mostly men of understanding, knowledge, jurisprudence, dialectics, and diverse doctrines.

2.2 Syria

ذِكْرُ الشَّامِ:

وَالشَّامُ هُوَ أَقْلِيمٌ وَاسِعٌ كَثِيرُ الْخَيْرِ، وَبِهِ دِمَشْقُ وَالرَّشِيدُ وَحِمَصٌ وَحَمَاةٌ وَحَلَبٌ وَأَنْطَاكِيَّةٌ وَاللَّادِقِيَّةُ وَيَبْتُ الْمَقْدِسِ وَالرَّمْلَةُ وَبَيْرُوتُ وَصُورُ وَصَيْدَاءُ، وَأَرْضُهُ صَالِحَةٌ لِلزَّرْعِ وَالضَّرْعِ، وَفِيهِ مِنَ الْجِبَالِ وَالْأَنْهَارِ وَالْعُيُونِ مَا لَا يُحْصَى، وَأَشْهُرُ أَنْهَارِهِ الْأُرْنُونَ وَاللَّتَانِي وَالْيَرْمُوكُ، وَالنَّاسُ فِيهِ أَهْلُ فِقْهِ وَعِلْمٍ وَصِحَّةٍ وَجَمَالٍ.

Syria is a wide clime, rich in blessings. In it are Damascus, al-Rashid, Homs, Hama, Aleppo, Antioch, Latakia, Jerusalem, al-Ramla, Beirut, Tyre, and Sidon. Its land is suitable for cultivation and pasturage; it contains innumerable mountains, rivers, and springs. Its most famous rivers are the Orontes, the Litani, and the Yarmuk. Its people are distinguished in jurisprudence, knowledge, health, and beauty.

3 The Eastern Climes

3.1 Khurasan and Transoxiana

ذِكْرُ خُرَاسَانَ وَمَا وَرَاءَ النَّهْرِ:

وَخُرَاسَانُ أَعْظَمُ أَقَالِيمِ الْمَشْرِقِ وَأَكْثَرُهَا بِلَادًا وَأَجْلَهَا مُلُوكًا، وَبِهَا نِيشَابُورُ وَمَرْوُ الشَّاهِجَانُ وَهَرَاةٌ وَبَلْخُ وَطَخَارِسْتَانُ وَبَلْخُ وَغُرْجَانُ وَطُوسُ وَسَابُورُ وَاسْتَخَرُ وَنَسَا وَابُيُورْدُ وَسَرَخْسُ وَمِيمَنْدَجَانُ وَخَوَارِزْمُ وَالسَّغْدُ وَالشَّاشُ وَفَارَابُ، وَهِيَ أَرْضٌ وَاسِعَةٌ كَثِيرَةُ الْمَاءِ وَالْبُسْتَانِ، وَأَكْثَرُهَا أَهْلُ عِلْمٍ وَأَدَبٍ وَتِجَارَةٍ.

Khurasan is the greatest of the eastern climes, with the most lands and the most splendid kings. In it are Nishapur, Merv the Great, Herat, Balkh, Tukharistan, Bushanj, Gurjan, Tus, Sabur, Istakhr, Nasa, Abiward, Sarakhs, Maymandajan, Khwarizm, Sogdia, Shash, and Farab. It is a wide land, abundant in water and orchards; its people are mostly devoted to learning, literature, and commerce.

وَمَا وَرَاءَ النَّهْرِ هُوَ بَخَارَى وَسَمَرْقَنْدٌ وَخُجَنْدَةُ وَالشَّاشُ وَأَسَدَابَادُ وَرَامِتَنُ وَنَسَفٌ وَكَاشُ وَشَاشُ وَالْأَقُّ وَطَلْقَانُ وَشِيرَازُ وَخَوَارِزْمُ، وَهُوَ إِقْلِيمٌ وَاسِعٌ كَثِيرُ الْخَيْرَاتِ، وَأَهْلُهُ مَتَمَسِّكُونَ بِالْعِلْمِ وَالِدِّينَ، وَبِهِمْ مِنَ الشَّجَاعَةِ وَالْفُتُوَّةِ مَا لَا يُوصَفُ، وَمِنْهُمْ أَهْلُ الْحَدِيثِ وَالْفَقْهِ وَالْتَفْسِيرِ، وَهُمْ أَكْثَرُ النَّاسِ حِفْظًا وَاتِّقَانًا.

Transoxiana consists of Bukhara, Samarkand, Khujanda, Shash, Asadabad, Ramitan, Nasa, Kish, Shash, Ilak, Talakan, Shiraz, and Khwarizm. It is a wide clime, rich in blessings; its people hold fast to knowledge and religion, and possess courage and generosity beyond description. Among them are the scholars of Hadith, jurisprudence, and exegesis; they are the most assiduous of men in memorization and precision.

4 Geographical Notices

4.1 The Calculation of Distances

وَقَدْ أَحْصَيْنَا مَسَافَاتِ الْبِلَادِ بِالْفَرَاسِجِ وَالْمِيلِ، وَالْفَرَسِجُ يُسَاوِي ثَلَاثَةَ أَمْيَالٍ، وَالْمِيلُ أَلْفُ ذِرَاعٍ، وَالذِّرَاعُ أَرْبَعَةٌ وَعِشْرُونَ أَصْبُعًا، وَقَدْ حَسَبْنَا مَا بَيْنَ مَكَّةَ وَالْمَدِينَةِ فَوَجَدْنَاهُ أَرْبَعِمِائَةَ فَرَسِجٍ، وَمَا بَيْنَ الْبَصْرَةِ وَبَغْدَادَ أَرْبَعِمِائَةَ فَرَسِجٍ، وَمَا بَيْنَ بَغْدَادَ وَسَامَرَاءَ أَرْبَعِمِائَةَ فَرَسِجٍ، وَمَا بَيْنَ دِمَشْقَ وَالرَّحْبَةِ ثَلَاثُونَ فَرَسِجًا، وَمَا بَيْنَ الْبَصْرَةِ وَعُمَانَ أَرْبَعِمِائَةَ فَرَسِجٍ.

We have measured the distances between cities in parasangs and miles. A parasang equals three miles; a mile is one thousand cubits; a cubit is twenty-four fingerbreadths. We have calculated [the following]: between Mecca and Medina: 400 parasangs; between Basra and Baghdad: 400 parasangs; between Baghdad and Samarra: 400 parasangs; between Damascus and al-Rahba: 30 parasangs; between Basra and Oman: 400 parasangs.

4.2 The Coordinates of Principal Cities

City	Longitude	Latitude
Mecca	77° 20'	21° 30'
Medina	77° 00'	25° 00'
Baghdad	79° 30'	33° 20'
Basra	80° 00'	30° 10'
Damascus	73° 30'	33° 30'
Jerusalem	73° 00'	31° 50'
Alexandria	68° 00'	31° 10'
Nishapur	88° 00'	36° 10'
Merv	87° 00'	37° 20'
Bukhara	93° 00'	39° 40'
Samarkand	94° 00'	39° 40'

Note: These coordinates are given according to the system of the Arab geographers, measured in degrees east of the Canary Islands (the westernmost point of the inhabited world in the Ptolemaic tradition).

Index of Geographical Names

Name (Arabic)	Modern Identification
مَكَّة	Mecca
الْمَدِينَة	Medina
الْكُوفَة	Kufa
الْبَصْرَة	Basra
بَغْدَاد	Baghdad
وَاسِط	Wasit
دِمَشْق	Damascus
حَلَب	Aleppo
بَيْتُ الْمَقْدِس	Jerusalem
مِصْر	Egypt (Fustat/Cairo)
الْإِسْكَنْدَرِيَّة	Alexandria
الْقَيْرَوَان	Kairouan
نِشَابُور	Nishapur
مَرْو	Merv
بَلْخ	Balkh
بُخَارَى	Bukhara
سَمَرْقَنْد	Samarkand
خَوَارِزْم	Khwarizm (Khiva)

KITAB AL-FIHRIST

The Book Catalogue

A COMPREHENSIVE BIBLIOGRAPHIC ENCYCLOPEDIA OF ARABIC
LITERATURE AND SCHOLARSHIP FROM THE 10TH CENTURY

Edited with Notes (*Mit Anmerkungen Herausgegeben*)

by
Gustav Flügel

Dr. Theol. und Philos., Professor, Comthul, Ritter etc.

Completed after his death by
Dr. Johannes Roediger & Dr. August Mueller

Author:

Muḥammad ibn Ishāq Ibn al-Nadīm

Baghdad, c. 990 CE

LEIPZIG

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Preface to the Notes (*Vorwort zu den Anmerkungen*)

Der vorliegenden Bearbeitung der Anmerkungen Professor Fl"ugels zu dem Texte des Fihrist konnte f"ur den ersten Theil bis S. 172 fast nur die von der Hand des Verfassers selbst herr"uhrende Reinschrift zu Grunde gelegt werden; f"ur die sp"ateren B"ucher war eine "ahnliche Reinschrift vorhanden, welche bis zu dem Komma hinter vor Anmerkung 3 zu S. 280 reicht, ausserdem ein meist sehr sauberes, "uberall ohne Zweideutigkeit lesbares Brouillon; ferner gew"ahrte neben wenigen Bl"attern mit einzelnen Notizen der urspr"ungliche ziemlich knappe Entwurf der Anmerkungen, welcher vollst"andig benutzt werden konnte, sp"arliche Ausbeute; nach Vollendung des ersten Bandes wurden mir auch Fl"ugels Originalabschrift des Textes sowie die Excerpte desselben aus Ibn al-Kifti und Ibn Abi Useibi'a durch das dankbar anzuerkennende Vertrauen der K. Universit"atsbibliothek zu Leipzig zug"anglich.

Dem mir gewordenen Auftrage gem"ass habe ich dies unfertige und auch, soweit es von Fl"ugel in der Reinschrift hinterlassen worden ist, immerhin der letzten Ueberarbeitung entbehrende Material, wo es noch n"othig erschien, einer neuen Redaction unterzogen. Ich bin dabei bestrebt gewesen, einerseits die hie und da vorkommenden Widerspr"uche und Wiederholungen zu beseitigen, den bisweilen etwas dunkeln Stil leichter verst"andlich zu machen, "uberhaupt durch sorgf"altige und einheitliche Behandlung aller Aeusserlichkeiten die Lesbarkeit des unvollendeten Werkes zu f"ordern. Es durfte indes der deutlich ausgepr"agte Charakter der Fl"ugelschen Arbeit nicht verwischt werden, ich habe daher nicht nur die von ihm gew"ahlte Transcription durchgef"uhrt und seine Ausdrucksweise m"oglichst geschont, sondern insbesondere fast aller materieller Aenderungen und Zus"atze mich enthalten, welche nicht durch die angedeuteten R"ucksichten durchaus geboten erschien, oder sich auf Versehen bezogen, die Fl"ugel jedenfalls selbst beseitigt haben w"urde.

Nur war es nach den von der Bearbeitung des ersten Bandes befolgten, mit des Verfassers eigenem Verfahren "ubereinstimmenden Grunds"atzen othwendig, hie und da die anf"angliche Fassung einer Anmerkung mit dem inzwischen weiter verbesserten Texte auszugleichen; doch ist auch dabei mit grosser Schonung verfahren worden, und ich habe es in manchen F"allen vorgezogen, auf einen bestehenden Widerspruch nur aufmerksam zu machen. Diese wie andere sparsame Zus"atze sind mit Ausnahme einiger litterarischer Nachweise und gelegentlicher Bezugnahme auf andere Stellen des Buches selbst in eckige Klammern gesetzt und mit *M.* (bez. *R.*) bezeichnet.

Halle, 24. Mai 1872.

August M"uller.

I (Vorwort zum ersten Bande)

Am 5. Juli 1870 entriss ein f"ur Fernerstehende unerwarteter Tod nach schweren Leiden **Gustav Fl"ugel** seiner trauernden Familie und setzte seiner rastlosen, dem Dienste der Wissenschaft geweihten Th"atigkeit ein leider zu fr"uhes Ziel. Es sollte ihm nicht verg"onnt sein das Werk selbst zu vollenden, welches ihn mehr als f"unfundzwanzig Jahre seines Lebens vorzugsweise besch"aftigt hatte und mit welchem sein Name schon so eng verkn"upft war. Er mag geahnt haben, dass die Tage seines Lebens gez"ahlt seien, als er mit erneutem Eifer die Herausgabe des Fihrist betrieb, und schon hatten sechs Bogen die Presse verlassen, als ihn der Tod abrief. So musste er es anderen "uberlassen zu Ende zu f"uhren was er begonnen; aber er hatte so weit vorgearbeitet, dass der Druck nach seinem Tode ohne wesentliche Unterbrechung fortgef"uhrt werden konnte.

Auf Antrag des Sohnes des Verstorbenen, des Herrn Assessor Dr. Fl"ugel und des Verlegers "ubernahm ich die ehrenvolle Aufgabe, die weitere Herausgabe des Fihrist zu besorgen, und kaum nach Jahresfrist erscheint jetzt in dem vorliegenden Bande der vollst"andige Text. Mit ihm schliesst meine specielle Th"atigkeit an dem Buche ab, da Herr Dr. August M"uller in Halle die Bearbeitung der Anmerkungen und die Anfertigung des Index, zu denen ich mich gleichfalls verpflichtet hatte, bereitwilligst "ubernommen hat.

Der *Text* lag druckfertig von Fl"ugels Hand vor, und es blieb mir nur "ubrig f"ur die correcte Drucklegung desselben Sorge zu tragen. Da er mir nur bogenweise zu Gesichte kam, so wurde, wenn der Druck nicht bedeutende Verz"ogerung erleiden sollte, eine gleichm"assige Durcharbeitung desselben mir sehr erschwert, und ich musste mich meist darauf beschr"anken kleine Versehen und Incorrectheiten, wie solche ja erst im Drucke recht in die Augen springen und sicher auch von dem verstorbenen Herausgeber bemerkt worden w"aren, zu berichtigen.

Breslau, im September 1871.

Dr. Johannes Roediger.

nd Its Author

Ibn al-Nadīm, whose full name is Abū al-Faraj Muḥammad ibn Ishāq ibn Abī Yaqūb al-Nadīm al-Warrāq al-Baghdādī (died c. 995/998 CE), was a Muslim scholar, bibliographer, and bookseller who lived in Baghdad during the Buyid period. His *Kitāb al-Fihrist* (كتاب الفهرست) is the most important bibliographic work in Arabic literature and an indispensable source for the history of Islamic civilization, especially for the period down to the 10th century.

1.1 The Author

Ibn al-Nadīm was a *warrāq* (book copier and seller) by profession, a trade that gave him unique access to the vast literary output of his time. His *nisba* “al-Nadīm” (النديم) suggests he may have been a companion or boon-companion, while “al-Warrāq” (الوراق) refers to his profession as a stationer. He moved in the intellectual circles of Baghdad and had personal knowledge of many of the scholars and authors whose works he catalogued.

The exact date of his death is uncertain. Internal evidence from the *Fihrist* suggests he was still working on it in 987/988 CE (377 AH), and he is generally believed to have died around 995–998 CE.

1.2 The *Kitāb al-Fihrist*

The *Fihrist* is organized into ten *discourses* (*maqālāt*, مقالات), covering the entirety of Arabic book culture:

1. Languages, scripts, and calligraphy – الفن الأول في الكتب واللغات والخطوط
2. Theology and sacred law – الفن الثاني في أصول الدين والفرق
3. Law and government – الفن الثالث في الفقه والمذاهب والقضاة والحكام
4. Scholarship and philology – الفن الرابع في النحو واللغة والتاريخ والعروض والأدب
5. Poetry and poets – الفن الخامس في الشعراء وأخبارهم
6. Religions, sects, and doctrines – الفن السادس في أصحاب الملل والنحل والأديان
7. Philosophy, mathematics, and natural sciences – الفن السابع في الفلسفة والحكمة والعلوم الطبيعية والرياضية
8. Stories, romances, and magic – الفن الثامن في الخرافات والأساطير والسحر
9. Books of the non-Muslim peoples – الفن التاسع في كتب السابئين والمجوس والنصارى والمشرّكين
10. History, biography, and genealogy – الفن العاشر في التاريخ والسير والأنساب

1.3 The Flügel Edition

This edition was prepared by Gustav Leberecht Flügel (1802–1870), the foremost German Arabist of his generation and professor at Leipzig. Flügel worked on the edition for more than 25 years but died before its completion. The work was finished by Dr. Johannes Roediger (Volume I: Text, published

st Müller (Volume II: Notes and Indices, published 1872), with the support of the *Deutsche Morgenländische Gesellschaft*.

The edition is based on multiple manuscripts designated by sigla:

- C. (= Codex) — The primary manuscript
- P. — Paris manuscript
- H. — Oxford (Bodleian) manuscript
- V. — Vienna manuscript
- L. — Leipzig manuscript

ic Entries on Mathematicians and Scientists

2.1 Chapter VII: Philosophy and the Sciences

The seventh discourse of the *Fihrist* is devoted to philosophy (الفلسفة), wisdom (الحكمة), the natural sciences (العلوم الطبيعية), and the mathematical sciences (العلوم الرياضية). This section preserves invaluable information about Greek, Persian, and Indian scientific works translated into Arabic, as well as original contributions by Muslim scholars.

2.1.1 The Translation Movement

Ibn al-Nadīm provides detailed accounts of the translators who rendered Greek scientific works into Arabic during the 8th–10th centuries:

Translator	Principal Works
حنين بن إسحاق	Ḥunayn ibn Ishāq (d. 873): The most prolific translator of Galen and Hippocrates; also translated geometrical and astronomical works.
ثابت بن قرة	Thābit ibn Qurra (d. 901): Mathematician, astronomer, and translator of Apollonius and Archimedes.
إسحاق بن حنين	Ishāq ibn Ḥunayn: Translated Euclid's <i>Elements</i> and Ptolemy's <i>Almagest</i> .
قسطا بن لوقا	Qusṭā ibn Lūqā (d. 912): Translated works on astronomy, mechanics, and geometry.
أبو عثمان الدمشقي	Abū Ṭhman al-Dimashqī: Translated Aristotle's logical and physical works.

2.1.2 Mathematical Sciences (العلوم الرياضية)

The *Fihrist* provides detailed catalogues of mathematical works:

وكتاب وكتاب أبولونيوس في المخروطات أربعة مقالات كتاب أقليدس في الأصول الهندسية عشرة مقالات وكتاب المجسطي لبطليموس وكتاب الخوارزمي في الجبر والمقابلة أرشميدس في الكرات والأسطوانات

[“The book of Euclid on geometrical foundations, ten treatises; the book of Apollonius on conic sections, four treatises; the book of Archimedes on spheres and cylinders; the book of al-Khwārizmī on algebra and al-muqābala; the *Almagest* of Ptolemy.” — Ed.]

2.1.3 Astronomy and Astrology (الهيئة والنجوم)

Scholar	Notable Works Listed
الخوارزمي	al-Khwārizmī: <i>Zīj al-Sindhīd</i> , astronomical tables; works on the astrolabe.

Scholar	Notable Works Listed
البتاني	al-Battānī (Albategnius, d. 929): <i>al-Zīj al-Ṣābī</i> , revised planetary theory.
أبو معشر	Abū Maṣār al-Balkhī (d. 886): <i>Kitāb al-Madkhal al-Kabīr</i> on astrology.
الفرغاني	al-Farghānī: <i>Kitāb fī al-Hay'a</i> , a summary of Ptolemaic astronomy.
إبراهيم بن سنان	Ibrāhīm ibn Sinān (d. 946): Works on the quadrature of the parabola and astronomical observations.

2.1.4 Medicine and Natural Philosophy

The *Fihrist* preserves detailed lists of medical works:

وبولس وبطليموس وغيرهم ومن أطباء الهند شاناق وأما أطباء اليونان فمنهم أبقرط وأرسطوطاليس وجالينوس وجبرائيل بن بختيشوع والحسن بن ثوبة وغيرهم وكاناك وغيرهم ومن أطباء الفرس يوحنا بن ماسويه وشرك

[“As for the Greek physicians, among them are Hippocrates, Aristotle, Galen, Paulus, Ptolemy, and others; among the Indian physicians: *Ṣānaq*, *Shark*, *Kānak*, and others; among the Persian physicians: *Yūḥannā ibn Māsawayh*, *ḡibrā’īl ibn Bakhtīshūc*, *al-Ḥasan ibn Thawāba*, and others.” — Ed.]

ries from the *Anmerkungen*

The following entries are representative samples from Flügel's commentary (*Anmerkungen*), illustrating the philological method and the interweaving of German scholarly apparatus with Arabic quotations.

3.1 On the Translation of Scientific Terms

Anm. 1. Hier schieben alle Codd., P. H. V., den bezuglosen Zusatz ein أَطال الله بقاء السيد الفاضل, eine bekannte hier an falschen Ort gerathene Phrase.

Anm. 3. الشعيرة wechselt im Arabischen mit الشعرة von شعره = شَعْدَه, Formen, die *Vullers* und der t"urkische K"am"us trotz des *ayn* f"ur urspr"unglich persisch gelten lassen.

Anm. 4. Vgl. Kor'an 26, 189 (und 7, 170), nicht يوم الظلمة wie in Hist. anteisl. S. 30 Z. 7; dagegen bei Ibn Kuteiba wiederholt يوم أخزى genannt (z. B. S. 21), da eine Wolke, als Schu'eib zu den Kindern Madjan geschickt wurde, Feuer auf die Ungl"aubigen regnete und diese vertilgte.

3.2 On Manuscript Variants

Anm. 5. Alle Codd. بن أد, dagegen sp"ater (s. Z. 21) P. V. بن اول, das eine mit eben so viel Recht als das andere. Bei Ab'l'fidā (Hist. anteisl. ed. *Fleischer* S. 192) heisst es:

ثم ولد للبسط ابن اده ثم ولد لابن اده ابنه اد ابن عثمان بن أد ونمیل

--- Tāschkōprizāda (Wien. Cod. Nr. 16 Bl. 20 und 165): عدنان بن أد; und *Ibn Kuteiba* S. 30: عدنان بن أد.

3.3 On Philosophers and Their Works

Anm. 10. أبو علي الفارسي; s. Ibn Chall. Nr. 162; Ann. Musl. II, S. 552 und 562 und Anm. 403; 64 Abdall. S. 481 (29); Anthol. gramm. 38 und 190 (31); *Hamaker Spec.* S. 43 (163); die gr. Schulen S. 110.

Anm. 11. الرواسي; s. die gr. Sch. S. 118 und hier 34, 18 Anm. 14. بن أبي سارة, wof"ur H. Chalfa بن حسن besetzt. سارة ist Frauenname; s. Ibn Kut. S. 38 und 115.

Anm. 12. معاذ الهزاز; s. Ibn Chall. Nr. 735; Ann. Musl. II, S. 86; die gr. Schulen S. 119.

ive Arabic Text Samples

4.1 On Philosophical Schools and Doctrines

The following passage from the *Fihrist* (Chapter VI/VII) discusses the transmission of Greek philosophy to the Islamic world:

في نفسه وله همة ومحبة للعلوم خطر بباله كان خالد بن يزيد بن معاوية يُسمّى حكيم آل مروان وكان فاضلاً بالعربية وأمرهم بنقل جماعة من فلاسفة اليونانيين ممن كان ينزل مدينة مصر وقد تفقه الصنعة فأمر بإحضار لغة ثم والقبطي إلى العربي وهذا أول نقل كان في الإسلام من لغة إلى الكتب في الصنعة من اللسان اليوناني نقل الديوان وكان باللغة الفارسية

[“Khālīd ibn Yazīd ibn Muḥāwiya, called the sage of the Marwānids, was virtuous, ambitious, and fond of the sciences. It occurred to him to commission a group of Greek philosophers residing in Egypt to translate books on the craft from Greek and Coptic into Arabic. This was the first translation in Islam from one language to another; then the dīwān was translated from Persian.” — Regarding the legendary role of Khālīd ibn Yazīd in initiating the translation movement.]

4.2 On Mathematical and Astronomical Works

وكان اليهوديوسائل في ابن محمد بن صول الكاتب أحد البلغاء والشعراء الفحصاء أبو إسحاق إبراهيم بن العباس السلاطين لما نبأ قال أبو تمام لولا أن همة إبراهيم سمّت به إلى خدمة مدينة جماعة من الخلفاء وكان طريفاً كتاب رسائل كتاب العطر الدولة كبير كتاب الطب كتاب ترك لشاعر خبراً يعني لجودة شعره وله من الكتب كتاب النوادر الصيد

[On Abū Ishāq Ibrāhīm ibn al-ḥAbbās ibn Muḥammad ibn Šul al-Kātib, one of the eloquent speakers and excellent poets, and his listed works including Kitāb al-Rasā'il, Kitāb al-ḥAṭr, Kitāb al-Dawla al-Kabīr, Kitāb al-Ṭibb, and Kitāb al-Nawādir.]

4.3 On Greek Philosophers Translated into Arabic

أبو جعفر محمد بن أحمد بن يحيى بن الأشعري ابن الحسن بن سعيد وله من الكتب كتاب الاحتجاجات زيدان كتاب ما نزل الشيعة والروايات والفقه وله من الكتب كتاب الجامع ويحتوي على عمران الأشعري من علماء كتاب النوادر كتاب ما أنزل الله في أمير المؤمنين من القرآن في الحسين وله من الكتب كتاب البيان

[Catalogue of the works of Abū Jaʿfar Muḥammad ibn Aḥmad ibn Yaḥyā ibn ʿImrān al-Ashʿarī, a scholar of Shīʿism, traditions, and jurisprudence, including Kitāb al-Jāmiʿ, Kitāb al-Bayān, and Kitāb al-Nawādir.]

Scholarly Apparatus

The second volume of the Flügel edition concludes with extensive indices that constitute an invaluable research tool:

5.1 Index of Personal Names (*Verzeichniss der Eigennamen*)

The index comprises approximately six printed quires (*Bogen*) and was in part excerpted by Flügel and Dr. Rödiger. As Müller notes:

Das Verzeichniss umfasst nur die Eigennamen von Personen und Stämmen, alles andere musste ausgeschlossen bleiben um den ohnehin dieser Abschnitte nicht noch zu vergrössern. Die dreifache Theilung ist aus Gründen der Uebersichtlichkeit erfolgt, obwol die Grenzen zwischen den verschiedenen Classen nicht überall scharf gezogen werden können. Die Namen sind angeführt, wie sie an den einzelnen Stellen sich finden, doch habe ich die Citate, welche sich auf dieselbe Persönlichkeit bezogen, möglichst zusammengebracht und dann durch Verweisungen jenem Grundsatzes genügt.

5.2 Sigla and Manuscripts

The following manuscript sigla are used throughout the edition:

Siglum	Location	Description
C.	Leiden	Primary codex, the basis of the edition
P.	Paris	Bibliothèque Nationale manuscript
H.	Oxford	Bodleian Library manuscript
V.	Vienna	Kaiserliche Hofbibliothek manuscript
L.	Leipzig	Universitätsbibliothek manuscript

5.3 Abbreviations

Abkürzung	Bedeutung
Codd.	Codices (Handschriften)
Vgl.	Vergleiche (compare)
Ann.	Anmerkung (note)
S.	Seite (page)
Z.	Zeile (line)
Bl.	Blatt (folio)
Ed.	Edition
Ueb.	Übersetzung (translation)

ignificance of the *Fihrist*

The *Kitāb al-Fihrist* is not merely a bibliography; it is a window onto the intellectual world of medieval Islam. For historians of mathematics and science, it provides:

1. **The most complete record of the Greek-to-Arabic translation movement** (8th–10th centuries), listing hundreds of works translated from Greek, Syriac, Persian, Sanskrit, and other languages.
2. **Detailed biographical information** about translators, scholars, and scientists, many of whom are not known from any other source.
3. **Systematic catalogues of scientific works** organized by discipline, showing the scope and structure of each field.
4. **Preservation of titles and incipits** of works now lost, including many Greek works that do not survive in the original.
5. **Evidence for the circulation of scientific texts** in 10th-century Baghdad, including information about which works were considered authoritative.

6.1 Key Figures in Mathematics and Astronomy

The *Fihrist* preserves information about the following important mathematical and astronomical authors and their works:

Name	Date	Significance
al-Khwārizmī	d. 850	Father of algebra; systematic treatment of linear and quadratic equations.
Thābit ibn Qurra	d. 901	Translated and extended Greek mathematics; work on amicable numbers.
Abū Kāmil	d. 930	<i>Algebra</i> ; systematic use of irrational numbers.
Ibrāhīm ibn Sinān	d. 946	Quadrature of the parabola; criticism of Ptolemaic astronomy.
al-Battānī	d. 929	<i>Zīj al-Ṣābī</i> : refined trigonometric methods.
Abū al-Wafā'	d. 998	Introduced tangent function; spherical trigonometry.

IBN AL-QIFṬĪ'S
TA'RĪḥ AL-ḤUKAMĀ'

AUF GRUND DER VORARBEITEN AUG. MÜLLER'S
HERAUSGEGEBEN

VON

Prof. Dr. Julius Lippert,

LEHRER AM SEMINAR FÜR ORIENTALISCHE SPRACHEN IN BERLIN.

MIT UNTERSTÜTZUNG DER KGL. AKADEMIE DER WISSENSCHAFTEN ZU BERLIN.

LEIPZIG,
DIETERICH'SCHE VERLAGSBUCHHANDLUNG
(THEODOR WEICHER)
1903.

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Vorwort

Zweck dieser Zeilen ist es zunächst, den Antheil festzustellen, den Aug. Müller an dem Zustandekommen dieser Ausgabe gehabt hat. Müller hat, nachdem er sich zur Edition des Werkes entschlossen, die Berliner Hs. 10053 mit gewohnter Sorgfalt copiert und dann die ihm erreichbaren anderen Hss., soweit sie ihm der Berücksichtigung werth erschienen, collationiert. Bei der grossen Zahl der in Betracht kommenden Mss. gebührt der Uebersichtlichkeit der Collation alles Lob. Auch die in (dem damals noch ungedruckten) Ibn Abi Uṣaibī'a sich findenden Parallelstellen hatte er schon am Rande seiner Abschrift hin und da notirt. Mir blieb also auf Grund des vorliegenden Materials nur noch die kritische Arbeit der Text herstellung übrig, die ich mit Zuhülfenahme der Berliner Hs. (Cod. or. qu. 786), die Müller noch nicht bekannt war, und der beiden Kairensen Hss. bewerkstelligte. Müller's Verdienst ist es also in erster Linie, wenn der *“Ta'rīḥ al-Ḥukamā”* das Licht der Welt erblickt hat.

Sodann möchte ich an dieser Stelle derer gedenken, die mich bei der Fertigstellung der Edition freundlichst unterstützt haben. Herr Prof. Dr. Aug. Fischer in Leipzig hat sich der Mühe unterzogen, die Correctur der drei ersten Bogen, Herr Privatdocent Dr. Horovitz in Berlin die der ersten Hälfte, Herr Dr. E. Mittwoch in Berlin im Anschluss daran die der zweiten Hälfte zu lesen. Mein hochverehrter Lehrer, Herr Geheimrath

Sachau, der von Anbeginn diese Arbeit mit bekanntem Wohlwollen gefördert, hat auch die Güte gehabt, die sämtlichen Correcturen dieses Werkes zu lesen. Seine zahlreichen Rathschläge und Fingerzeige haben manches Dunkel erhält und sind mir auch für die Text herstellung von wesentlichem Nutzen gewesen. Nach erfolgter Drucklegung hatte noch Herr Ahmed Zeki Bey in Kairo die Freundlichkeit, den ganzen Text noch einmal einer Durchsicht zu unterziehen; seine werthvollen Noten machen einen beträchtlichen Theil der Emendanda aus. Herr Dr. Kern in Berlin hat mir die Benutzung und Beurtheilung der beiden Kairensen Hss. durch die Abschrift und Collation der umfangreichen Muḥtār vīta ermöglicht, Herr Privatdocent Dr. J. Friedlaender in Strassburg mehrfache, die dortige Hs. betreffende Anfragen bereitwilligst beantwortet. Allen diesen Herren sage ich meinen tiefgefühltesten Dank für ihre mir bereitwilligst gewährte Unterstützung.

Mein Dank gebührt aber auch der Deutschen Morgenländischen Gesellschaft für die jahrelange Ueberlassung der Müller'schen Vorarbeiten, sowie insbesondere der Hohen Akademie der Wissenschaften in Berlin, deren liberale Munificenz durch Gewährung eines beträchtlichen Kostenzuschusses erst die Drucklegung ermöglicht hat.

Introduction to the Work

Jamāl al-Dīn Abū al-Ḥasan ‘Alī ibn Yūsuf ibn Ibrāhīm ibn ‘Abd al-Wāḥid al-Shaybānī al-Qiftī (1172–1248 CE) was an Egyptian Arab scholar, historian, and Ayyubid vizier. Born in Qift (Coptos) in Upper Egypt, he studied in Cairo before moving to Jerusalem and later Aleppo, where he compiled most of his works. Of the twenty-six works known by title, only two survive: *Inbāh al-Ruwāt ‘alā Anbāh al-Nuḥāt* (containing some thousand biographies of Muslim scholars) and the present work.

The *تاريخ الحكماء* (*Tārīkh al-Ḥukamā’*, “History of the Learned”) is a biographical dictionary of 414 physicians, philosophers, astronomers, and scientists — including Greeks, Indians, Persians, and Arabs. The text preserved today is an epitome made in 1249/647 AH by *محمد بن علي الخطيب الزوزني* (Muḥammad ibn ‘Alī al-Khaṭībī al-Zawzanī), a year after Ibn al-Qiftī’s death, from the longer original titled *كتاب إخبار العلماء بأخبار الحكماء* (*Kitāb Ikḥār al-Ulamā’ bi-Akḥbār al-Ḥukamā’*, “Notices of Scholars with the Accounts of Learned Men”).

The present edition was prepared by **Prof. Dr. Julius Lippert** (1839–1909), building upon the preparatory work of August Müller. It was published in Leipzig by Dieterich’sche Verlagsbuchhandlung in 1903, with the support of the Royal Academy of Sciences in Berlin. Lippert collated the Berlin manuscript (Cod. or. quart. 786) against two Cairene manuscripts and employed the then-unpublished *‘Uyūn al-Anbā’* of Ibn Abī Uṣaybī’a for parallel readings.

Structure and Content

The work is arranged alphabetically by the names of its subjects. It opens with the author’s introduction, followed by biographical entries ranging from the ancient Greek philosophers and physicians (beginning with Hermes Trismegistus and Asclepius) through the Hellenistic, Byzantine, and Islamic periods, down to scholars contemporary with the author. Each entry typically provides: the subject’s full name and genealogical information (*nisba*), dates of birth and death, teachers and students, works composed, and noteworthy anecdotes.

Mathematical and Scientific Figures

Among the 414 biographies, notable entries include:

- **Hermes Trismegistus** (هرمس) — the legendary patron of science
- **Aesculapius** (أبو قراط) — the Greek god of medicine
- **Pythagoras** (فيثاغورس) — mathematician and philosopher
- **Socrates, Plato, Aristotle** — the Athenian philosophers
- **Euclid** (أقليدس) — the geometer
- **Galen** (جالينوس) — the physician
- **Ṭabarī, Rāzī, Ibn Sīnā** (Avicenna) — Islamic polymaths
- **Bīrūnī, Khayyām, Tūsī** — Central Asian scientists

The Arabic Title Page

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

الْحُكَمَاءُ بِأَخْبَارِ الْعُلَمَاءِ إِنْخَبَارِ كِتَابُ

الْقَفْطِيِّ الشَّيْبَانِيِّ الْوَاحِدِ عَبْدِ بْنِ إِبْرَاهِيمَ بْنِ يُوسُفَ بْنِ عَلِيٍّ الْحَسَنِ أَبِي الدِّينِ الْجَمَالِ

الْخَطِيبِ عَلِيِّ بْنِ مُحَمَّدٍ وَهُوَ الزَّوْزَنِيُّ مُخْتَصَرٌ

Im Namen Gottes, des Barmherzigen, des Gnädigen. Das Buch der Nachrichten der Gelehrten über die Geschichten der Weisen, von Jamāl al-Dīn Abī al-Ḥasan ‘Alī ibn Yūsuf ibn Ibrāhīm ibn ‘Abd al-Wāḥid al-Shaybānī al-Qifṭī. Die Epitome des al-Zawzanī, und das ist Muḥammad ibn ‘Alī al-Khaṭīb.

1 Author's Introduction

قَدْ عَزَمْتُ — بِتَأْيِيدِ اللَّهِ — عَلَى ذِكْرِ مَنْ اشتهر ذكره من الحكماء من كل قبيلة وأمة، قديمها وحديثها، إلى زمانِي، وما حفظ عنه من قول انفرد به، أو كتاب صنفه، أو حكمة عليه ابتدعتها، ونسبت إليه، فإنني رأيت ذلك من الأمور التي جهلت والتواريخ التي هجرت، وفي مطالعة هذا اعتبار بمن مضى، وذكر من خلف، وهو اعتبار أرجو به الثواب لي ولقارئه إن شاء الله تعالى، وقد قفيت له ليسهل تناوله، والله الموفق.

English Translation:

I have resolved — with God's aid — to mention those among the learned men whose fame has become celebrated, from every tribe and nation, ancient and modern, down to my own time; and to record what has been preserved from them of unique sayings, books they have composed, or sublime wisdom they have originated and which has been attributed to them. For I have seen that this is among matters that have been neglected and histories that have been abandoned; and in perusing this there is instruction by those who have passed and mention of those who come after. And this is instruction through which I hope for reward for myself and for its reader, God Most High willing. I have arranged it alphabetically so that it may be easily consulted. God is the One who grants success.

German Translation:

Ich habe beschlossen — mit Gottes Hilfe — diejenigen unter den Weisen zu erwähnen, deren Ruhm berühmt geworden ist, aus jedem Stamm und jeder Nation, alt und neu, bis zu meiner Zeit; und aufzuzeichnen, was von ihnen überliefert worden ist an einzigartigen Aussprüchen, Büchern, die sie verfasst haben, oder erhabener Weisheit, die sie geschaffen haben und die ihnen zugeschrieben wurde. Denn ich habe gesehen, dass dies zu den vernachlässigten Angelegenheiten und den verlassenen Geschichten gehört; und in der Lektüre davon liegt Belehrung durch die Vergangenen und Erwähnung der Nachkommenden. Und dies ist eine Belehrung, durch die ich Belohnung für mich und für seinen Leser erhoffe, so Gott der Erhabene will. Ich habe es alphabetisch geordnet, damit es leicht konsultiert werden kann. Gott ist der Erfolg gebende.

2 Selected Biographical Entries

The following entries represent a selection from the 414 biographies contained in the work, arranged alphabetically. The Arabic text is presented with critical apparatus (footnotes) following Lippert's edition. The page references in footnotes refer to the 1903 edition pagination.

2.1 Entry 1: Ibn Riḍwān al-Miṣrī (ابن رضوان المصري)

ابن رِضْوَانُ الْمِصْرِيِّ
فَتَلَدَ هَذَا وَمَعَ الْهَيْئَةِ وَلَا الْمَنْظَرَ حَسَنٌ يَكُنْ وَلَمْ الْحَقِيقِينَ لَا الْمُعَلِّقِينَ مِنْ وَكَانَ الْمَنْطِقِ مِنْ وَشَيْئًا الطِّبِّ مِنْ شَيْئًا قَرَأَ ثُمَّ
مُبْتَكِرَةً! مُلْتَقَطَةً مُخْتَصَرَةً هِيَ بَلْ بَيَانُهَا غَايَةٌ فِي تَكُنْ لَمْ كُتِبَ وَصَنَّفَ ذِكْرُهُ وَسَارَ عَنْهُ وَأَخَذُوا الطُّلُبَةَ مِنْ جَمَاعَةٍ لَهُ

Translation: “Ibn Riḍwān the Egyptian. Then he read something of medicine and something of logic, and he was among the commentators (*mu'alliqīn*), not the investigators (*muḥaqqiqīn*). He was not handsome in appearance or demeanour; yet despite this, a group of students formed an attachment to him, learned from him, and his fame spread. He composed books that were not in the highest style of eloquence, but rather were abbreviated, compiled, and novel!”¹

2.2 Entry 2: Abū 'Alī ibn Sīnā (Avicenna) (أبو علي بن سينا)

أَبُو عَلِيٍّ بْنُ سِينَا
فِيهَا: قَصِيدَةٌ هُنَاكَ وَأَنْشَاءٌ فَرَدَجَانٍ، لَهَا فَلَبَعَةٍ إِلَى فَاتُوهُ فَأَخَذُوهُ أَعْدَائِهِ بَعْضٌ عَلَيْهِ ذِمْلًا طَلَبَتْهُ فِي وَجَدَ فَلَهَا
التَّخْرِيجَ أَمْرٍ فِي الشَّكِّ وَكُلُّ تَرَاهُ كَمَا بِالتَّيَقِينِ دُخِلَتْ
رَجَعَ ثُمَّ بَعْضُهَا الْقَلْعَةَ تِلْكَ إِلَى وَمَرَّ الْمَلِكُ نَائِجٌ عَنْهَا وَانْهَزَمَ وَأَخَذَهَا هَمْدَانُ، الدَّوْلَةُ عَلَاءٌ قَصَدَ ثُمَّ أَشْهُرُ أَرْبَعَةٍ فِيهَا وَبَقِيَ
دَارٍ فِي وَتَزَلَ هَمْدَانُ، إِلَى الشَّيْخِ مَعَهُمْ وَحَمَلُوا هَمْدَانُ، إِلَى الدَّوْلَةِ شَمْسٍ وَأَبْنُ الْمَلِكِ نَائِجٌ وَهَادَ هَمْدَانُ، عَنْ الدَّوْلَةِ عَلَاءُ
الْشِّفَاءِ. كِتَابٌ مِنَ الْمَنْطِقِ بِتَصْنِيفٍ هُنَاكَ وَاشْتَغَلَ الْعُلُو

Translation: “When some of his enemies found fault with him in his quest, they seized him and brought him to a fortress with two towers, and there he composed a poem which includes the line:

‘I entered with certainty as you see it, / And all doubt in the matter of graduation’

He remained there four months, then headed for ‘Alā’ al-Dawla in Hamadān, and took it; the protégé of the king was routed from it. Then some of them passed by that fortress, then ‘Alā’ al-Dawla returned from Hamadān, and the protégé of the king and Ibn Shams

¹This entry, though brief, is characteristic of Ibn al-Qifti's candid biographical style. Ibn Riḍwān (998–1068) was a prominent Egyptian physician and philosopher, known for his commentary on Galen's *Ars Parva* and his polemics against ‘Alī ibn Riḍwān's medical theories. The phrase “*mu'alliqīn*, not *muḥaqqiqīn*” suggests that he was a compiler rather than an original investigator — a distinction that reflects the Aristotelian epistemological framework of Islamic scholarship.

al-Dawla went to Hamadān, and they carried the Shaykh with them to Hamadān, and he settled in the house of al-'Alwī and there occupied himself with composing the logic section of the *Kitāb al-Shifā'*.²

2.3 Entry 3: Pythagoras (فيثاغورس)

فيثاغورس
 فيثاغورس وكان سُقراط، ذلِكَ فِي وَتَلَاهُ الْإِلَهِيَّ، الْعِلْمُ فِي تَكَلَّمَ مِنْ أَوَّلُ — صَوْرَ أَهْلِ مِنْ وَهُوَ — الصَّمُودِيَّ فيثاغورس
 إِيطَالِيَا إِلَى مِنْهَا وَخَرَجَ صَوْرَ إِلَى رَجَعَ ثُمَّ سَنَةً، أَرْبَعِينَ ذلِكَ عَلَى وَبَقِيَ وَالْهَرَمَسِيَّةَ لِلْمُنَجِّمِينَ تَلَدَ أَنْ بَعْدَ مِصْرَ مِنْ خَرَجَ قَدْ
 بِهَا. وَأَقَامَ

Translation: “Pythagoras the Samian — he is from the people of Tyre — was the first to discourse on divine science (*al-ilm al-ilāhī*), and Socrates followed him in that. Pythagoras had left Egypt after studying with the astrologers and the Hermetists, and remained at that for forty years; then he returned to Tyre and left from there to Italy, where he settled.”³

2.4 Entry 4: Euclid (أقليدس)

أقليدس
 وَهُوَ «فَنَدْرُوسَا»، سَمَاهُ كِتَابًا الْمُنَجِّمِينَ فِي صَنَفَ قَدْ ذلِكَ قَبْلَ وَكَانَ مَشْهُورًا، كِتَابًا فِيهَا وَصَنَفَ بِالْهَيْئَةِ، عَالِمُ أَقْلِيدَسُ:
 كَثِيرَةً. أَشْيَاءَ عَنْهُ التَّرْجَمَانُ نَقَلَ وَقَدْ تَأَلَّفَ، إِنَّهُ يُقَالُ

Translation: “Euclid: a scholar in geometry (*hay'a*), and he composed on it a famous book. Before that he had composed a book on astrologers which he called *Phaenomena* (*Fanandrūsā*), which is said to be lost, and the translator has transmitted many things from it.”⁴

2.5 Entry 5: Yaḥyā al-Naḥwī (John of Nikiu) (يحيى النحوي)

يَحْيَى النَّحْوِيُّ الْمِصْرِيُّ الْإِسْكَندَرَانِيَّ، تَلْمِيزُ سَاوِيرُسَ، وَكَانَ أَسْقَفًا فِي كَنِيسَةِ الْإِسْكَندَرِيَّةِ بِمِصْرَ، وَكَانَ يَدْعُو إِلَى طَرِيقَةِ
 الْيَعْقُوبِيَّةِ مِنَ النَّصْرَى.

Translation: “Yaḥyā al-Naḥwī the Egyptian, the Alexandrian, disciple of Severus, and he was a bishop in the church of Alexandria in Egypt, and he advocated the Jacobite way of the Christians.”⁵

²Ibn Sīnā (980–1037 CE) is the most celebrated philosopher-physician in the Islamic tradition. The *Kitāb al-Shifā'* (Book of Healing) is his encyclopedic masterwork, encompassing logic, natural philosophy, mathematics, and metaphysics. The verse quoted plays on the double meaning of *takhrīj* — “graduation” from medical study and “deduction” in logic.

³This entry reflects the standard Arabic account of Pythagoras's intellectual biography, which includes the Egypt sojourn as a foundational element of Greek philosophy. The identification of Pythagoras as a student of the “Hermetists” (*al-Haramisiyya*) connects Greek wisdom to the Egyptian Hermetic tradition.

⁴Euclid's *Phaenomena* (*Φαινόμενα*) is an early work of spherical astronomy, concerned with the risings and settings of celestial bodies. It survives in the Greek tradition and was indeed known in Arabic translation.

⁵This is the famous entry in which Ibn al-Qiftī preserves the account of the destruction of the Library of

3 The Name Index (أَسْمَاءُ الْأَشْخَاصِ)

The edition includes a comprehensive Arabic index of personal names (pp. 38–76 of the printed edition), arranged alphabetically. The following is a representative sample from the letter أ (alif), illustrating the index format:

Page	Name	Entry
2, s.	آخِرُ أَفْلَاطُون (The Last Plato)	—
11, s.	أَبُقْرَاط (Hippocrates)	—
51, s.	أَبُو بَكْرٍ الرَّازِيّ (al-Rāzī)	—
83, s.	أَبُو الْحُسَيْنِ بْنِ سِينَا (Ibn Sīnā)	Avicenna
64, s.	أَبُو نَعِيمٍ الْأَصْبَهَانِيّ	—
390, s.	أَرِسْطُوطَالِيس (Aristotle)	—
2, s. 11, s. 314, s.	أَفْلَاطُون (Plato)	—
397, s.	أَنَا كُنُوش (Anaxagoras?)	—
158, s.	أَنَا كَسْمَانَس (Anaximenes)	—
296, s.	أَنَا كَسْمَانْدَرَس (Anaximander)	—
424, s.	أَنْتُونِسُ بْنُ يَحْيَى النَّحْوِيّ	—

Alexandria under ‘Amr ibn al-‘Āṣ. The biography continues to narrate how al-Naḥwī was dismissed from his bishopric for rejecting the doctrine of the Trinity after reading philosophical books, and how he was received by ‘Amr ibn al-‘Āṣ, the Arab conqueror of Egypt. The story of the library’s destruction — with Caliph ‘Umar’s alleged command “If the books agree with the Qur’ān, they are superfluous; if they disagree, they are heretical” — is one of the most frequently cited (and debated) passages in the history of the book. See the English translation by Emily Cottrell in Roger Pearse, *Tertullian Project* (2017).

4 Editorial Apparatus

4.1 Sigla (Manuscript Sources)

Siglum	Description
A	Berlin, Königliche Bibliothek, Cod. or. quart. 786 (Neu collationiert)
B	Berlin, Königliche Bibliothek, Cod. or. oct. 10053 (Müller's Vorlage)
C	Kairo, Biblothèque Nationale (alte Sammlung), Ms. I
D	Kairo, Biblothèque Nationale (alte Sammlung), Ms. II
V	Wien, Kaiserliche Hofbibliothek, Cod. Mixt. 783
M	München, Königliche Hof- und Staatsbibliothek, Cod. ar. 651

4.2 Critical Symbols

[] — Einfügung des Herausgebers (editorial insertion)

< > — Auslassung oder Unleserlichkeit (lacuna or illegibility)

a), b), c) — Kritische Anmerkungen am Seitenfuss (critical footnotes)

BC, A, etc. — Varianten nach den Handschriften (manuscript variants)

4.3 Note on the Arabic Text

The Arabic text in this edition follows the standard conventions of 19th-century Orientalist printing. The *hamza* and diacritical points are supplied where the manuscripts are ambiguous or defective. The editorial footnotes indicate manuscript variants using the sigla listed above, and emendations are marked with square brackets. The pagination of the original edition is indicated in the margins.

End of specimen edition

THE HINDU-ARABIC NUMERALS

BY
DAVID EUGENE SMITH
AND
LOUIS CHARLES KARPINSKI

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AND LOUIS CHARLES KARPINSKI

Preface

So familiar are we with the numerals that bear the misleading name of Arabic, and so extensive is their use in Europe and the Americas, that it is difficult for us to realize that their general acceptance in the transactions of commerce is a matter of only the last four centuries, and that they are unknown to a very large part of the human race to-day.

It seems strange that such a labor-saving device should have struggled for nearly a thousand years after its system of place value was perfected before it replaced such crude notations as the one that the Roman conqueror made substantially universal in Europe. Such, however, is the case, and there is probably no one who has not at least some slight passing interest in the story of this struggle. To the mathematician and the student of civilization the interest is generally a deep one; to the teacher of the elements of knowledge the interest may be less marked, but nevertheless it is real; and even the business man who makes daily use of the curious symbols by which we express the numbers of commerce, cannot fail to have some appreciation for the story of the rise and progress of these tools of his trade.

This story has often been told in part, but it is a long time since any effort has been made to bring together the fragmentary narrations and to set forth the general problem of the origin and development of these numerals. In this little work we have attempted to state the history of these forms in small compass, to place before the student materials for the investigation of the problems involved, and to express as clearly as possible the results of the labors of scholars who have studied the subject in different parts of the world. We have had no theory to exploit, for the history of mathematics has seen too much of this tendency already, but as far as possible we have weighed the testimony and have set forth what seem to be the reasonable conclusions from the evidence at hand.

To facilitate the work of students an index has been prepared which we hope may be serviceable. In this the names of authors appear only when some use has been made of their opinions or when their works are first mentioned in full in a footnote.

If this work shall show more clearly the value of our number system, and shall make the study of mathematics seem more real to the teacher and student, and shall offer material for interesting some pupil more fully in his work with numbers, the authors will feel that the considerable labor involved in its preparation has not been in vain.

We desire to acknowledge our especial indebtedness to Professor Alexander Ziwet for reading all the proof, as well as for the digest of a Russian work, to Professor Clarence L. Meader for Sanskrit transliterations, and to Mr. Steven T. Byington for Arabic transliterations and the scheme of pronunciation of Oriental names, and also our indebtedness to other scholars in Oriental learning for information.

DAVID EUGENE SMITH
LOUIS CHARLES KARPINSKI

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Pronunciation of Oriental Names

(S) = in Sanskrit names and words; (A) = in Arabic names and words.

b, d, f, g, h, j, l, m, n, p, sh (A), t, th (A), v, w, x, z, as in English.

a, (S) like *u* in *but*: thus *pandit*, pronounced *pundit*. (A) like *a* in *ask* or in *man*.

a with macron, as in *father*.

c, (S) like *ch* in *church* (Italian *c* in *cento*).

d with dot below, n with dot below, s with dot below, t with dot below, (S) made with the tip of the tongue turned up and back into the dome of the palate.

d with dot below, s with dot below, t with dot below, z with dot below, (A) made with the tongue spread so that the sounds are produced largely against the side teeth.

d with dot below (A), like *th* in *this*.

e, (S) as in *they*. (A) as in *bed*.

g with dot below, (A) a voiced consonant formed below the vocal cords; its sound is compared by some to a *g*, by others to a guttural *r*; in Arabic words adopted into English it is represented by *gh* (e.g. *ghoul*), less often *r* (e.g. *razzia*).

h preceded by b, c, t, etc. does not form a single sound with these letters, but is a more or less distinct *h* sound following them; cf. the sounds in *abhor*, *boathook*, etc., or, more accurately for (S), the *bhoys* etc. of Irish brogue.

h (A) retains its consonant sound at the end of a word.

h with dot below, (A) an unvoiced consonant formed below the vocal cords; its sound is sometimes compared to German hard *ch*, and may be represented by an *h* as strong as possible.

i, as in *pin*. **i with macron**, as in *pique*.

k, as in *kick*.

kh, (A) the hard *ch* of Scotch *loch*, German *ach*, especially of German as pronounced by the Swiss.

m, n, (S) like French final *m* or *n*, nasalizing the preceding vowel.

n with dot below, see *d with dot below*. **n with dot above**, like *ng* in *singing*.

o, (S) as in *so*. (A) as in *obey*.

q, (A) like *k* (or *c*) in *cook*; further back in the mouth than in *kick*.

r, (S) English *r*, smooth and untrilled. (A) stronger. **r with dot below**, (S) used as vowel, as in *apron* when pronounced *aprn* and not *apera*; modern Hindus say *ri*, hence our *amrita*, *Krishna*, for *a-mrta*, *Krsna*.

s with acute, (S) English *sh* (German *sch*).

s, as in *same*. **s with dot below**, see *d with dot below*.

t, see *d with dot below*.

u, as in *put*. **u with macron**, as in *rule*.

y, as in *you*.

z, see *d with dot below*.

ayin / hamza ('), (A) a sound kindred to the spiritus lenis (that is, to our ears, the mere distinct separation of a vowel from the preceding sound, as at the beginning of a word in German) and to *h with dot below*. The ' is a very distinct sound in Arabic, but is more nearly represented by the spiritus lenis than by any sound that we can produce without much special training. That is, it should be treated as silent, but the sounds that precede and follow it should not run together. In Arabic words adopted into English it is treated as silent, e.g. in *Arab*, *amber*, *Caaba*.

Accent: (S) as if Latin; (A) on the last syllable that contains a long vowel or a vowel followed by two consonants, except that a final long vowel is not ordinarily accented; if there is no long vowel nor two consecutive consonants, the accent falls on the first syllable. The words *al* and *ibn* are never accented.

1 Early Ideas of Their Origin

It has long been recognized that the common numerals used in daily life are of comparatively recent origin. The number of systems of notation employed before the Christian era was about the same as the number of written languages, and in some cases a single language had several systems. The Egyptians, for example, had three systems of writing, with a numerical notation for each; the Greeks had two well-defined sets of numerals, and the Roman symbols for number changed more or less from century to century. Even to-day the number of methods of expressing numerical concepts is much greater than one would believe before making a study of the subject, for the idea that our common numerals are universal is far from being correct.

It will be well, then, to think of the numerals that we still commonly call Arabic, as only one of many systems in use just before the Christian era. As it then existed the system was no better than many others, it was of late origin, it contained no zero, it was cumbersome and little used, and it had no particular promise. Not until centuries later did the system have any standing in the world of business and science; and had the place value which now characterizes it, and which requires a zero, been worked out in Greece, we might have been using Greek numerals to-day instead of the ones with which we are familiar.

Of the first number forms that the world used this is not the place to speak. Many of them are interesting, but none had much scientific value. In Europe the invention of notation was generally assigned to the eastern shores of the Mediterranean until the critical period of about a century ago,—sometimes to the Hebrews, sometimes to the Egyptians, but more often to the early trading Phoenicians.¹

The idea that our common numerals are Arabic in origin is not an old one. The mediaeval and Renaissance writers generally recognized them as Indian, and many of them expressly stated that they were of Hindu origin. Others argued that they were probably invented by the Chaldeans or the Jews because they increased in value from right to left, an argument that would apply quite as well to the Roman and Greek systems, or to any other. It was, indeed, to the general idea of notation that many of these writers referred, as is evident from the words of England's earliest arithmetical textbook-maker, Robert Recorde (c. 1542):

“In that thinge all men do agree, that the Chaldays, whiche fyrste inuented thys arte, did set these figures as thei set all their letters. for they wryte backwarde as you tearme it, and so doo they reade. And that may appeare in all Hebrewe, Chaldaye and Arabike bookes...where as the Greekes, Latines, and all nations of Europe, do wryte and reade from the lefte hand towarde the ryghte.”²

Others, and among them such influential writers as Tartaglia in Italy and Kobel in Germany, asserted the Arabic origin of the numerals, while still others left the matter undecided or simply dismissed them as “barbaric.” Of course the Arabs themselves never laid claim to the invention, always recognizing their indebtedness to the Hindus both for the numeral forms and for the distinguishing feature of place value.

Foremost among these writers was the great master of the golden age of Bagdad, one of the first of the Arab writers to collect the mathematical classics of both the East and the West, preserving them

¹Discipulus. Quis primus invenit numerum apud Hebraeos et Aegyptios? Magister. Abraham primus invenit numerum apud Hebraeos, deinde Moses; et Abraham tradidit istam scientiam numeri ad Aegyptios, et docuit eos: deinde Josephus.” [Bede, *De computo dialogus* (doubtfully assigned to him), Opera omnia, Paris, 1802, Vol. I, p. 650.] “Alii referunt ad Phoenices inventores arithmetices, propter eandem commerciorum causam: Alii ad Indos: Ioannes de Sacrobosco, cujus sepulchrum est Lutetiae in comitio Maturinensi, refert ad Arabes.” [Ramus, *Arithmeticae libri duo*, Basel, 1569, p. 112.]

²From the 1558 edition of *The Ground of Artes*, fol. C, 5. Similarly Bishop Tonstall writes: “Numerandi artem a Chaldaeis esse profectam: qui dum scribunt, a dextra incipiunt, et in leuam progrediuntur.” [*De arte supputandi*, London, 1522, fol. B, 3.] Gemma Frisius, the great continental rival of Recorde, had the same idea: “Primum autem appellamus dexterum locum, eo quod haec ars vel a Chaldaeis, vel ab Hebraeis ortum habere credatur, qui etiam eo ordine scribunt”; but this refers more evidently to the Arabic numerals. [*Arithmeticae practicae methodus facilis*, Antwerp, 1540, fol. 4 of the 1563 ed.]

and finally passing them on to awakening Europe. This man was Mohammed the Son of Moses, from Khwarezm, or, more after the manner of the Arab, **Mohammed ibn Musa al-Khwarazmi**,³ a man of great learning and one to whom the world is much indebted for its present knowledge of algebra and of arithmetic.

Of him there will often be occasion to speak; and in the arithmetic which he wrote, and of which Adelhard of Bath (c. 1130) may have made the translation or paraphrase, he stated distinctly that the numerals were due to the Hindus. This is as plainly asserted by later Arab writers, even to the present day. Indeed the phrase *'ilm hindi*, "Indian science," is used by them for arithmetic, as also the adjective *hindi* alone.

Probably the most striking testimony from Arabic sources is that given by the Arabic traveler and scholar **Mohammed ibn Ahmed, Abu 'l-Rihan al-Biruni** (973–1048), who spent many years in Hindustan. He wrote a large work on India, the "Book of the Ciphers," unfortunately lost, which treated doubtless of the Hindu art of calculating, and was the author of numerous other works. Al-Biruni was a man of unusual attainments, being versed in Arabic, Persian, Sanskrit, Hebrew, and Syriac, as well as in astronomy, chronology, and mathematics. In his work on India he gives detailed information concerning the language and customs of the people of that country, and states explicitly that the Hindus of his time did not use the letters of their alphabet for numerical notation, as the Arabs did. He also states that the numeral signs called *ahka*⁴ had different shapes in various parts of India, as was the case with the letters. In his *Chronology of Ancient Nations* he gives the sum of a geometric progression and shows how, in order to avoid any possibility of error, the number may be expressed in three different systems: with Indian symbols, in sexagesimal notation, and by an alphabet system which will be touched upon later. He also speaks of "179, 876, 755, expressed in Indian ciphers," thus again attributing these forms to Hindu sources.

Preceding Al-Biruni there was another Arabic writer of the tenth century, Motahhar ibn Tahir, author of the Book of the Creation and of History, who gave as a curiosity, in Indian (Nagari) symbols, a large number asserted by the people of India to represent the duration of the world.

Mention should also be made of a widely-traveled student, **Al-Mas'udi** (885?–956), whose journeys carried him from Bagdad to Persia, India, Ceylon, and even across the China sea. He states that the wise men of India, assembled by the king, composed the *Sindhind*. Further on he states, upon the authority of the historian Mohammed ibn 'Ali 'Abdi, that by order of Al-Mansur many works of science and astrology were translated into Arabic, notably the *Sindhind* (Siddhanta). Concerning the meaning and spelling of this name there is considerable diversity of opinion. Colebrooke first pointed out the connection between Siddhanta and Sindhind. He ascribes to the word the meaning "the revolving ages."

This Sindhind is the book, says Mas'udi, which gives all that the Hindus know of the spheres, the stars, arithmetic, and the other branches of science. He mentions also Al-Khwarazmi and Habash as translators of the tables of the Sindhind. Al-Biruni refers to two other translations from a work furnished by a Hindu who came to Bagdad as a member of the political mission which Sindh sent to the caliph Al-Mansur, in the year of the Hejira 154 (a.d. 771).

The oldest work, in any sense complete, on the history of Arabic literature and history is the *Kitab al-Fihrist*, written in the year 987 a.d., by Ibn Abi Ya'qub al-Nadim. It is of fundamental importance for the history of Arabic culture. Of the ten chief divisions of the work, the seventh demands attention in this discussion for the reason that its second subdivision treats of mathematicians and astronomers.

The first of the Arabic writers mentioned is Al-Kindi (800–870 A.D.), who wrote five books on arithmetic and four books on the use of the Indian method of reckoning. Sened ibn 'Ali, the Jew, who was converted to Islam under the caliph Al-Mamun, is also given as the author of a work on the Hindu

³His full name was Abu 'Abdallah Mohammed ibn Musa al-Khwarazmi. He was born in Khwarezm, "the lowlands," the country about the present Khiva and bordering on the Oxus, and lived at Bagdad under the caliph al-Mamun. He died probably between 220 and 230 of the Mohammedan era, that is, between 835 and 845 a.d., although some put the date as early as 812.

⁴The Hindu name for the symbols of the decimal place system.

method of reckoning. To Al-Sufi, who died in 986 a.d., is also credited a large work on the same subject, and similar treatises by other writers are mentioned. We are therefore forced to the conclusion that the Arabs from the early ninth century on fully recognized the Hindu origin of the new numerals.

Leonard of Pisa, of whom we shall speak at length in the chapter on the Introduction of the Numerals into Europe, wrote his *Liber Abbaci* in 1202. In this work he refers frequently to the nine Indian figures, thus showing again the general consensus of opinion in the Middle Ages that the numerals were of Hindu origin. Some interest also attaches to the oldest documents on arithmetic in our own language. One of the earliest treatises on algorism is a commentary on a set of verses called the *Carmen de Algorismo*, written by Alexander de Villa Dei (Alexandre de Ville-Dieu), a Minorite monk of about 1240 a.d. The text of the first few lines is as follows:

*Haec algorismus ars praesens dicitur, in qua
Talibus Indorum fruimur bis quinque figuris.*

“This boke is called the boke of algorim or augrym after lewder use. And this boke tretys of the Craft of Nombryng, the quych crafte is called also Algorym. Ther was a kyng of Inde the quich heyth Algor and he made this craft. Algorisms, in the quych we use teen figurys of Inde.”

2 Early Hindu Forms with No Place Value

While it is generally conceded that the scientific development of astronomy among the Hindus towards the beginning of the Christian era rested upon Greek or Chinese sources, yet their ancient literature testifies to a high state of civilization, and to a considerable advance in sciences, in philosophy, and along literary lines, long before the golden age of Greece. From the earliest times even up to the present day the Hindu has been wont to put his thought into rhythmic form. The first of this poetry—it well deserves this name, being also worthy from a metaphysical point of view—consists of the Vedas, hymns of praise and poems of worship, collected during the Vedic period which dates from approximately 2000 B.C. to 1400 B.C.

Following this work, or possibly contemporary with it, is the Brahmanic literature, which is partly ritualistic (the Brahmanas), and partly philosophical (the Upanishads). Our especial interest is in the Sutras, versified abridgments of the ritual and of ceremonial rules, which contain considerable geometric material used in connection with altar construction, and also numerous examples of rational numbers the sum of whose squares is also a square, i.e. Pythagorean numbers, although this was long before Pythagoras lived.

Whitney places the whole of the Veda literature, including the Vedas, the Brahmanas, and the Sutras, between 1500 B.C. and 800 B.C., thus agreeing with Birk who holds that the knowledge of the Pythagorean theorem revealed in the Sutras goes back to the eighth century B.C. The importance of the Sutras as showing an independent origin of Hindu geometry, contrary to the opinion long held by Cantor of a Greek origin, has been repeatedly emphasized in recent literature, especially since the appearance of the important work of Von Schroeder.

Further fundamental mathematical notions such as the conception of irrationals and the use of gnomons, as well as the philosophical doctrine of the transmigration of souls,—all of these having long been attributed to the Greeks,—are shown in these works to be native to India. Although this discussion does not bear directly upon the origin of our numerals, yet it is highly pertinent as showing the aptitude of the Hindu for mathematical and mental work, a fact further attested by the independent development of the drama and of epic and lyric poetry.

It should be stated definitely at the outset, however, that we are not at all sure that the most ancient forms of the numerals commonly known as Arabic had their origin in India. As will presently be seen, their forms may have been suggested by those used in Egypt, or in Eastern Persia, or in China, or on the plains of Mesopotamia. We are quite in the dark as to these early steps; but as to their development in

India, the approximate period of the rise of their essential feature of place value, their introduction into the Arab civilization, and their spread to the West, we have more or less definite information. When, therefore, we consider the rise of the numerals in the land of the Sindhu, it must be understood that it is only the large movement that is meant, and that there must further be considered the numerous possible sources outside of India itself and long anterior to the first prominent appearance of the number symbols.

No one attempts to examine any detail in the history of ancient India without being struck with the great dearth of reliable material. So little sympathy have the people with any save those of their own caste that a general literature is wholly lacking, and it is only in the observations of strangers that any all-round view of scientific progress is to be found.

There is evidence that primary schools existed in earliest times, and of the seventy-two recognized sciences writing and arithmetic were the most prized. In the Vedic period, say from 2000 to 1400 B.C., there was the same attention to astronomy that was found in the earlier civilizations of Babylon, China, and Egypt, a fact attested by the Vedas themselves. Such advance in science presupposes a fair knowledge of calculation, but of the manner of calculating we are quite ignorant and probably always shall be.

One of the Buddhist sacred books, the *Lalitavistara*, relates that when the Bodhisattva was of age to marry, the father of Gopa, his intended bride, demanded an examination of the five hundred suitors, the subjects including arithmetic, writing, the lute, and archery. Having vanquished his rivals in all else, he is matched against Arjuna the great arithmetician and is asked to express numbers greater than 100 kotis.⁵ In reply he gave a scheme of number names as high as 10^{53} , adding that he could proceed as far as 10^{421} , all of which suggests the system of Archimedes and the unsettled question of the indebtedness of the West to the East in the realm of ancient mathematics.

Sir Edwin Arnold, in *The Light of Asia*, does not mention this part of the contest, but he speaks of Buddha's training at the hands of the learned Visvamitra:

And Viswamitra said, 'It is enough,
Let us to numbers. After me repeat
Your numeration till we reach the lakh,⁶
One, two, three, four, to ten, and then by tens
To hundreds, thousands.' After him the child
Named digits, decads, centuries, nor paused,
The round lakh reached, but softly murmured on,
Then comes the koti, nahut, ninahut,
Khamba, viskhamba, abab, attata,
To kumuds, gundhikas, and utpalas,
By pundarikas into padumas,
Which last is how you count the utmost grains
Of Hastagiri ground to finest dust;
But beyond that a numeration is,
The Katha, used to count the stars of night,
The Koti-Katha, for the ocean drops;
Ingga, the calculus of circulars;
Sarvanikchepe, by the which you deal
With all the sands of Gunga, till we come
To Antah-Kalpas, where the unit is
The sands of the ten crore Gungas. If one seeks

⁵I.e. $100 \cdot 10^7$.

⁶I.e. to 100,000. The lakh is still the common large unit in India, like the myriad in ancient Greece and the million in the West.

More comprehensive scale, th' arithmic mounts
 By the Asankya, which is the tale
 Of all the drops that in ten thousand years
 Would fall on all the worlds by daily rain;
 Thence unto Maha Kalpas, by the which
 The gods compute their future and their past.'

Thereupon Visvamisra Acarya⁷ expresses his approval of the task, and asks to hear the "measure of the earth" as far as yojana, the longest measure bearing name. This given, Buddha adds:

'And master! if it please,
 I shall recite how many sun-motes lie
 From end to end within a yojana.'
 Thereat, with instant skill, the little prince
 Pronounced the total of the atoms true.
 But Viswamisra heard it on his face
 Prostrate before the boy;
 'Art Teacher of thy teachers—thou, not I,
 For thou,' he cried, 'Art Guru.'

It is needless to say that this is far from being history. And yet it puts in charming rhythm only what the ancient Lalitavistara relates of the number-series of the Buddha's time. While it extends beyond all reason, nevertheless it reveals a condition that would have been impossible unless arithmetic had attained a considerable degree of advancement.

To this pre-Christian period belong also the Vedāṅgas, or "limbs for supporting the Veda," part of that great branch of Hindu literature known as Smṛiti (recollection), that which was to be handed down by tradition. Of these the sixth is known as Jyotiṣa (astronomy), a short treatise of only thirty-six verses, written not earlier than 300 B.C., and affording us some knowledge of the extent of number work in that period.

The Hindus also speak of eighteen ancient Siddhāntas or astronomical works, which, though mostly lost, confirm this evidence. As to authentic histories, however, there exist in India none relating to the period before the Mohammedan era (622 a.d.). About all that we know of the earlier civilization is what we glean from the two great epics, the Mahābhārata and the Rāmāyana, from coins, and from a few inscriptions.

It is with this unsatisfactory material, then, that we have to deal in searching for the early history of the Hindu-Arabic numerals, and the fact that many unsolved problems exist and will continue to exist is no longer strange when we consider the conditions. It is rather surprising that so much has been discovered within a century, than that we are so uncertain as to origins and dates and the early spread of the system.

The early Hindu numerals may be classified into three great groups, (1) the Kharosthi, (2) the Brahmi, and (3) the word and letter forms; and these will be considered in order.

The Kharosthi numerals are found in inscriptions formerly known as Bactrian, Indo-Bactrian, and Aryan, and appearing in ancient Gandhara, now eastern Afghanistan and northern Punjab. The alphabet of the language is found in inscriptions dating from the fourth century B.C. to the third century A.D., and from the fact that the words are written from right to left it is assumed to be of Semitic origin. No numerals, however, have been found in the earliest of these inscriptions, number-names probably having been written out in words as was the custom with many ancient peoples.

Not until the time of the powerful King Asoka, in the third century B.C., do numerals appear in any inscriptions thus far discovered; and then only in the primitive form of marks, quite as they would be

⁷I.e. the Wise.

found in Egypt, Greece, Rome, or in various other parts of the world. These Asoka inscriptions, some thirty in all, are found in widely separated parts of India, often on columns, and are in the various vernaculars that were familiar to the people. Two are in the Kharosthi characters, and the rest in some form of Brahmi. In the Kharosthi inscriptions only four numerals have been found, and these are merely vertical marks for one, two, four, and five.

In the so-called Saka inscriptions, possibly of the first century B.C., more numerals are found, and in more highly developed form, the right-to-left system appearing, together with evidences of three different scales of counting,—four, ten, and twenty.

There are several noteworthy points to be observed in studying this system. In the first place, it is probably not as early as that shown in the Nana Ghat forms hereafter given, although the inscriptions themselves at Nana Ghat are later than those of the Asoka period. The four is to this system what the X was to the Roman, probably a canceling of three marks as a workman does to-day for five, or a laying of one stick across three others. The ten has never been satisfactorily explained. It is similar to the A of the Kharosthi alphabet, but we have no knowledge as to why it was chosen. The twenty is evidently a ligature of two tens, and this in turn suggested a kind of radix, so that ninety was probably written in a way reminding one of the *quatre-vingt-dix* of the French.

This system has many points of similarity with the Nabatean numerals in use in the first centuries of the Christian era. The cross is here used for four, and the Kharosthi form is employed for twenty. In addition to this there is a trace of an analogous use of a scale of twenty. While the symbol for 100 is quite different, the method of forming the other hundreds is the same. The correspondence seems to be too marked to be wholly accidental.

It is not in the Kharosthi numerals, therefore, that we can hope to find the origin of those used by us, and we turn to the second of the Indian types, the Brahmi characters. The alphabet attributed to Brahma is the oldest of the several known in India, and was used from the earliest historic times.

The following numerals are, as far as known, the only ones to appear in the Asoka edicts. These fragments from the third century B.C., crude and unsatisfactory as they are, are the undoubted early forms from which our present system developed. They next appear in the second century B.C. in some inscriptions in the cave on the top of the Nana Ghat hill, about seventy-five miles from Poona in central India.

The cave was made as a resting-place for travelers ascending the hill, which lies on the road from Kalyana to Junar. It seems to have been cut out by a descendant of King Satavahana, for inside the wall opposite the entrance are representations of the members of his family, much defaced, but with the names still legible. It would seem that the excavation was made by order of a king named Vedisiri, and “the inscription contains a list of gifts made on the occasion of the performance of several yagnas or religious sacrifices,” and numerals are to be seen in no less than thirty places.

There is considerable dispute as to what numerals are really found in these inscriptions, owing to the difficulty of deciphering them; but the following, which have been copied from a rubbing, are probably number forms.

The next very noteworthy evidence of the numerals, and this quite complete as will be seen, is found in certain other cave inscriptions dating back to the first or second century A.D. In these, the Nasik cave inscriptions, the forms are as follows.

From this time on, until the decimal system finally adopted the first nine characters and replaced the rest of the Brahmi notation by adding the zero, the progress of these forms is well marked.

With respect to these numerals it should first be noted that no zero appears in the table, and as a matter of fact none existed in any of the cases cited. It was therefore impossible to have any place value, and the numbers like twenty, thirty, and other multiples of ten, one hundred, and so on, required separate symbols except where they were written out in words. The ancient Hindus had no less than twenty of these symbols, a number that was afterward greatly increased.

We have thus far spoken of the Kharosthi and Brahmi numerals, and it remains to mention the third

type, the word and letter forms. These are, however, so closely connected with the perfecting of the system by the invention of the zero that they are more appropriately considered in the next chapter, particularly as they have little relation to the problem of the origin of the forms known as the Arabic.

Having now examined types of the early forms it is appropriate to turn our attention to the question of their origin. As to the first three there is no question. The I or — is simply one stroke, or one stick laid down by the computer. The II represents two strokes or two sticks, and so for the III and IV. From some primitive form came the two of Egypt, of Rome, of early Greece, and of various other civilizations.

From some primitive form — came the Chinese symbol, which is practically identical with the symbols found commonly in India from 150 B.C. to 700 A.D. In the cursive form it becomes z, and this was frequently used for two in Germany until the 18th century. It finally went into the modern form 2, and the = in the same way became our 3.

There is, however, considerable ground for interesting speculation with respect to these first three numerals. The earliest Hindu forms were perpendicular. In the Nana Ghat inscriptions they are vertical. But long before either the Asoka or the Nana Ghat inscriptions the Chinese were using the horizontal forms for the first three numerals, but a vertical arrangement for four. Now where did China get these forms? Surely not from India, for she had them, as her monuments and literature show, long before the Hindus knew them.

The tradition is that China brought her civilization around the north of Tibet, from Mongolia, the primitive habitat being Mesopotamia, or possibly the oases of Turkestan. Now what numerals did Mesopotamia use? The Babylonian system, simple in its general principles but very complicated in many of its details, is now well known. In particular, one, two, and three were represented by vertical arrow-heads. Why, then, did the Chinese write theirs horizontally? The problem now takes a new interest when we find that these Babylonian forms were not the primitive ones of this region, but that the early Sumerian forms were horizontal.

What interpretation shall be given to these facts? Shall we say that it was mere accident that one people wrote "one" vertically and that another wrote it horizontally? This may be the case; but it may also be the case that the tribal migrations that ended in the Mongol invasion of China started from the Euphrates while yet the Sumerian civilization was prominent, or from some common source in Turkestan, and that they carried to the East the primitive numerals of their ancient home, the first three, these being all that the people as a whole knew or needed.

As to the numerals above three there have been very many conjectures. The figure one of the Demotic looks like the one of the Sanskrit, the two (reversed) like that of the Arabic, the four has some resemblance to that in the Nasik caves, the five (reversed) to that on the Ksatrapa coins, the nine to that of the Kusana inscriptions, and other points of similarity have been imagined.

More absurd is the hypothesis of a Greek origin, supposedly supported by derivation of the current symbols from the first nine letters of the Greek alphabet. This difficult feat is accomplished by twisting some of the letters, cutting off, adding on, and effecting other changes to make the letters fit the theory. This peculiar theory was first set up by Dasypodius (Conrad Rauhfuß), and was later elaborated by Huet.

Out of all these conflicting theories, and from all the resemblances seen or imagined between the numerals of the West and those of the East, what conclusions are we prepared to draw as the evidence now stands? Probably none that is satisfactory. Indeed, upon the evidence at hand we might properly feel that everything points to the numerals as being substantially indigenous to India. And why should this not be the case? If the king Srong-tsan-Gampo (639 a.d.), the founder of Lhasa, could have set about to devise a new alphabet for Tibet, and if the Siamese, and the Singhalese, and the Burmese, and other peoples in the East, could have created alphabets of their own, why should not the numerals also have been fashioned by some temple school, or some king, or some merchant guild?

We may summarize this chapter by saying that no one knows what suggested certain of the early numeral forms used in India. The origin of some is evident, but the origin of others will probably

never be known. There is no reason why they should not have been invented by some priest or teacher or guild, by the order of some king, or as part of the mysticism of some temple. Whatever the origin, they were no better than scores of other ancient systems and no better than the present Chinese system when written without the zero, and there would never have been any chance of their triumphal progress westward had it not been for this relatively new symbol.

3 Later Hindu Forms, with a Place Value

Before speaking of the perfected Hindu numerals with the zero and the place value, it is necessary to consider the third system mentioned above—the word and letter forms. The use of words with place value began at least as early as the 6th century of the Christian era. In many of the manuals of astronomy and mathematics, and often in other works in mentioning dates, numbers are represented by the names of certain objects or ideas.

For example, zero is represented by “heaven-space” or “the void” (*ambara* or *akasa*); one by “moon” (*sasi* or *indu*), “earth” (*bhu*), “beginning” (*adi*), “Brahma,” or, in general, by anything markedly unique; two by “eyes” (*nayana*) or “hands” (*kara*), “the twins” or “Yama”; four by “oceans,” five by “senses” or “arrows” (the five arrows of Kamadeva); six by “seasons” or “flavors”; seven by “mountain” (*ago*), and so on. These names, accommodating themselves to the verse in which scientific works were written, had the additional advantage of not admitting, as did the figures, easy alteration, since any change would tend to disturb the meter.

As an example of this system, the date “Saka Samvat, 867” (a.d. 945 or 940), is given by “*giri-rasa-vasu*” meaning “mountains” (seven), “the flavors” (six), and the gods of which there were eight. In reading the date these are read from right to left. The period of invention of this system is uncertain. The first trace seems to be in the Srautasutra of Katyayana and Latyayana. It was certainly known to Varaha-Mihira (d. 587), for he used it in the *Brhat-Samhita*.

Mention should also be made, in this connection, of a curious system of alphabetic numerals that sprang up in southern India. In this we have the numerals represented by the letters as given in the following table.

By this plan a numeral might be represented by any one of several letters, and thus it could the more easily be formed into a word for mnemonic purposes. For example, the word *kha gont yan me sa ma pa* has the value 1,565,132, reading from right to left. This, the oldest specimen (1184 a.d.) known of this notation, is given in a commentary on the Rigveda, representing the number of days that had elapsed from the beginning of the Kaliyuga.

A second system of this kind is still used in the pagination of manuscripts in Ceylon, Siam, and Burma, having also had its rise in southern India. In this the thirty-four consonants when followed by *a* (as *ka*, *la*) designate the numbers 1–34; by *a* (as *ka*, *la*), those from 35 to 68; by *i* (*ki*, *li*), those from 69 to 102, inclusive; and so on.

As already stated, however, the Hindu system as thus far described was no improvement upon many others of the ancients, such as those used by the Greeks and the Hebrews. Having no zero, it was impracticable to designate the tens, hundreds, and other units of higher order by the same symbols used for the units from one to nine. In other words, there was no possibility of place value without some further improvement. So the Nana Ghat symbols required the writing of “thousand seven twenty-four” about like 7, tw, 4 in modern symbols, instead of 7024, in which the seven of the thousands, the two of the tens (concealed in the word twenty, being originally “twain of tens,” the *-ty* signifying ten), and the four of the units are given as spoken and the order of the unit (tens, hundreds, etc.) is given by the place. To complete the system only the zero was needed; but it was probably eight centuries after the Nana Ghat inscriptions were cut, before this important symbol appeared; and not until a considerably later period did it become well known.

To enable us to appreciate the force of this argument a large number, 8,443,682,155, may be con-

sidered as the Hindus wrote and read it, and then, by way of contrast, as the Greeks and Arabs would have read it.

Modern American reading: 8 billion, 443 million, 682 thousand, 155.

Hindu: 8 padmas, 4 vyarbudas, 4 kotis, 3 prayutas, 6 laksas, 8 ayutas, 2 sahasra, 1 sata, 5 dasan, 5.

Arabic and early German: eight thousand thousand thousand and four hundred thousand thousand and forty-three thousand thousand, and six hundred thousand and eighty-two thousand and one hundred fifty-five (or five and fifty).

Greek: eighty-four myriads of myriads and four thousand three hundred sixty-eight myriads and two thousand and one hundred fifty-five.

As Woepcke pointed out, the reading of numbers of this kind shows that the notation adopted by the Hindus tended to bring out the place idea. No other language than the Sanskrit has made such consistent application, in numeration, of the decimal system of numbers. The introduction of myriads as in the Greek, and thousands as in Arabic and in modern numeration, is really a step away from a decimal scheme.

When the *ankapalli*, the decimal-place system of writing numbers, was perfected, the tenth symbol was called the *sunyabindu*, generally shortened to *sunya* (the void). Broekhaus has well said that if there was any invention for which the Hindus, by all their philosophy and religion, were well fitted, it was the invention of a symbol for zero. This making of nothingness the crux of a tremendous achievement was a step in complete harmony with the genius of the Hindu.

It is generally thought that this *sunya* as a symbol was not used before about 500 A.D., although some writers have placed it earlier. Since Aryabhata gives our common method of extracting roots, it would seem that he may have known a decimal notation, although he did not use the characters from which our numerals are derived. Moreover, he frequently speaks of the *void*.⁸ If he refers to a symbol this would put the zero as far back as 500 a.d., but of course he may have referred merely to the concept of nothingness.

A little later, but also in the sixth century, Varaha-Mihira wrote a work entitled *Brhat Samhita* in which he frequently uses *sunya* in speaking of numerals, so that it has been thought that he was referring to a definite symbol. This, of course, would add to the probability that Aryabhata was doing the same. It should also be mentioned as a matter of interest, and somewhat related to the question at issue, that Varaha-Mihira used the word-system with place value as explained above.

The first kind of alphabetic numerals and also the word-system (in both of which the place value is used) are plays upon, or variations of, position arithmetic, which would be most likely to occur in the country of its origin.

At the opening of the next century (c. 620 a.d.) Bana wrote of Subandhu's *Vasavadatta* as a celebrated work, and mentioned that the stars dotting the sky are here compared with zeros, these being points as in the modern Arabic system. On the other hand, a strong argument against any Hindu knowledge of the symbol zero at this time is the fact that about 700 A.D. the Arabs overran the province of Sind and thus had an opportunity of knowing the common methods used there for writing numbers. And yet, when they received the complete system in 776 they looked upon it as something new. Such evidence is not conclusive, but it tends to show that the complete system was probably not in common use in India at the beginning of the eighth century.

On the other hand, we must bear in mind the fact that a traveler in Germany in the year 1700 would probably have heard or seen nothing of decimal fractions, although these were perfected a century before that date. The elite of the mathematicians may have known the zero even in Aryabhata's time,

⁸Using *kha*, a synonym of *sunya*.

while the merchants and the common people may not have grasped the significance of the novelty until a long time after. On the whole, the evidence seems to point to the west coast of India as the region where the complete system was first seen.

Concerning the earliest epigraphical instances of the use of the nine symbols, plus the zero, with place value, there is some question. Colebrooke in 1807 warned against the possibility of forgery in many of the ancient copper-plate land grants. On this account Fleet, in the *Indian Antiquary*, discusses at length this phase of the work of the epigraphists in India, holding that many of these forgeries were made about the end of the eleventh century.

Buhler gives the copper-plate Gurjara inscription of Cedi-samvat 346 (595 a.d.) as the oldest epigraphical use of the numerals "in which the symbols correspond to the alphabet numerals of the period and the place." Vincent A. Smith quotes a stone inscription of 815 A.D., dated Samvat 872. So F. Kielhorn in the *Epigraphia Indica* gives a Pathari pillar inscription of Parabala, dated Vikrama-samvat 917, which corresponds to 861 A.D.

That there should be any inscriptions of date as early even as 750 A.D., would tend to show that the system was at least a century older. As will be shown in the further development, it was more than two centuries after the introduction of the numerals into Europe that they appeared there upon coins and inscriptions.

While Thibaut does not consider it necessary to quote any specific instances of the use of the numerals, he states that traces are found from 590 a.d. on. "That the system now in use by all civilized nations is of Hindu origin cannot be doubted; no other nation has any claim upon its discovery, especially since the references to the origin of the system which are found in the nations of western Asia point unanimously towards India."

The testimony and opinions of men like Buhler, Kielhorn, V. A. Smith, Bhandarkar, and Thibaut are entitled to the most serious consideration. As authorities on ancient Indian epigraphy no others rank higher. Their work is accepted by Indian scholars the world over, and their united judgment as to the rise of the system with a place value—that it took place in India as early as the sixth century a.d.—must stand unless new evidence of great weight can be submitted to the contrary.

4 The Symbol Zero

What has been said of the improved Hindu system with a place value does not touch directly the origin of a symbol for zero, although it assumes that such a symbol exists. The importance of such a sign, the fact that it is a prerequisite to a place-value system, and the further fact that without it the Hindu-Arabic numerals would never have dominated the computation system of the western world, make it proper to devote a chapter to its origin and history.

It was some centuries after the primitive Brahmi and Kharosthi numerals had made their appearance in India that the zero first appeared there, although such a character was used by the Babylonians in the centuries immediately preceding the Christian era. The symbol is known to us from their tablets, and apparently it was not used in calculation. Nor does it always occur when units of any order are lacking; thus 180 is written with the meaning three sixties and no units, since 181 immediately following is three sixties and one unit.

The earliest undoubted occurrence of a zero in India is an inscription at Gwalior, dated Samvat 933 (876 A.D.). Where 50 garlands are mentioned (line 20), 50 is written. The Bakshali Manuscript probably antedates this, using the point or dot as a zero symbol. Bayley mentions a grant of Jaika Rashtrakuta of Bharuj, found at Okamandel, of date 738 A.D., which contains a zero, and also a coin with indistinct Gupta date 707 (897 a.d.), but the reliability of Bayley's work is questioned.

Aside from its appearance in early inscriptions, there is still another indication of the Hindu origin of the symbol in the special treatment of the concept zero in the early works on arithmetic. Brahmagupta, who lived in Ujjain, the center of Indian astronomy, in the early part of the seventh century, gives in his

arithmetic a distinct treatment of the properties of zero. He does not discuss a symbol, but he shows by his treatment that in some way zero had acquired a special significance not found in the Greek or other ancient arithmetics. A still more scientific treatment is given by Bhaskara, although in one place he permits himself an unallowed liberty in dividing by zero. The most recently discovered work of ancient Indian mathematical lore, the *Ganita-Sara-Sangraha* of Mahaviracarya (c. 830 a.d.), while it does not use the numerals with place value, has a similar discussion of the calculation with zero.

What suggested the form for the zero is, of course, purely a matter of conjecture. The dot, which the Hindus used to fill up lacunae in their manuscripts, much as we indicate a break in a sentence, would have been a more natural symbol; and this is the one which the Hindus first used and which most Arabs use to-day. There was also used for this purpose a cross, like our X, and this is occasionally found as a zero symbol.

The small circle was possibly suggested by the spurred circle which was used for ten. It has also been thought that the omicron used by Ptolemy in his *Almagest*, to mark accidental blanks in the sexagesimal system which he employed, may have influenced the Indian writers. This symbol was used quite generally in Europe and Asia, and the Arabic astronomer Al-Battani (died 929 a.d.) used a similar symbol in connection with the alphabetic system of numerals.

Although the dot was used at first in India, as noted above, the small circle later replaced it and continues in use to this day. The Arabs, however, did not adopt the circle, since it bore some resemblance to the letter which expressed the number five in the alphabet system. The earliest Arabic zero known is the dot, used in a manuscript of 873 a.d. Sometimes both the dot and the circle are used in the same work, having the same meaning.

The Name of Zero

The name of this all-important symbol also demands some attention, especially as we are even yet quite undecided as to what to call it. We speak of it to-day as zero, naught, and even cipher; the telephone operator often calls it O, and the illiterate or careless person calls it aught. In view of all this uncertainty we may well inquire what it has been called in the past.

As already stated, the Hindus called it *sunya*, "void." This passed over into the Arabic as *as-sifr* or *sifr*. When Leonard of Pisa (1202) wrote upon the Hindu numerals he spoke of this character as *zephirum*. Maximus Planudes (1330), writing under both the Greek and the Arabic influence, called it *tziphra*.

Of course the English *cipher*, French *chiffre*, is derived from the same Arabic word, *as-sifr*, but in several languages it has come to mean the numeral figures in general. A trace of this appears in our word *ciphering*, meaning figuring or computing.

From *as-sifr* has come *zephyr*, *cipher*, and finally the abridged form *zero*. The earliest printed work in which is found this final form appears to be Calandri's arithmetic of 1491, while in manuscript it appears at least as early as the middle of the fourteenth century. It also appears in a work, *Le Kadran des marchans*, by Jehan Certain, written in 1485. This word soon became fairly well known in Spain and France.

The medieval writers also spoke of it as the *sipos*, and occasionally as the *galgal* or "wheel," *circulus*, *nota*, *theca*,⁹ *omicron* (the Greek o), being sometimes written *o* or ϕ to distinguish it from the letter *o*. It also went by the name *null* or *nulla*, and very commonly by the name *cipher*.

⁹Long supposed to be from its resemblance to the Greek theta, but explained by Petrus de Dacia as being derived from the name of the iron used to brand thieves and robbers with a circular mark placed on the forehead or on the cheek.

5 The Question of the Introduction of the Numerals into Europe by Boethius

Just as we were quite uncertain as to the origin of the numeral forms, so too are we uncertain as to the time and place of their introduction into Europe. There are two general theories as to this introduction. The first is that they were carried by the Moors to Spain in the eighth or ninth century, and thence were transmitted to Christian Europe, a theory which will be considered later. The second, advanced by Woepcke, is that they were not brought to Spain by the Moors, but that they were already in Spain when the Arabs arrived there, having reached the West through the Neo-Pythagoreans.

There are two facts to support this second theory: (1) the forms of these numerals are characteristic, differing materially from those which were brought by Leonardo of Pisa from Northern Africa early in the thirteenth century (before 1202 a.d.); (2) they are essentially those which tradition has so persistently assigned to Boethius (c. 500 A.D.), and which he would naturally have received, if at all, from these same Neo-Pythagoreans or from the sources from which they derived them.

Furthermore, Woepcke points out that the Arabs on entering Spain (711 A.D.) would naturally have followed their custom of adopting for the computation of taxes the numerical systems of the countries they conquered, so that the numerals brought from Spain to Italy, not having undergone the same modifications as those of the Eastern Arab empire, would have differed, as they certainly did, from those that came through Bagdad.

The theory is that the Hindu system, without the zero, early reached Alexandria (say 450 a.d.), and that the Neo-Pythagorean love for the mysterious and especially for the Oriental led to its use as something bizarre and cabalistic; that it was then passed along the Mediterranean, reaching Boethius in Athens or in Rome, and to the schools of Spain, being discovered in Africa and Spain by the Arabs even before they themselves knew the improved system with the place value.

A recent theory set forth by Bubnov also deserves mention, chiefly because of the seriousness of purpose shown by this well-known writer. Bubnov holds that the forms first found in Europe are derived from ancient symbols used on the abacus, but that the zero is of Hindu origin. This theory does not seem tenable, however, in the light of the evidence already set forth.

Two questions are presented by Woepcke's theory: (1) What was the nature of these Spanish numerals, and how were they made known to Italy? (2) Did Boethius know them?

The Spanish forms of the numerals were called the *huruf al-gobar*, the *gobar* or dust numerals, as distinguished from the *huruf al-jumal* or alphabetic numerals. Probably the latter, under the influence of the Syrians or Jews, were also used by the Arabs. The significance of the term *gobar* is doubtless that these numerals were written on the dust abacus, this plan being distinct from the counter method of representing numbers. It is also worthy of note that Al-Biruni states that the Hindus often performed numerical computations in the sand.

The gober numerals themselves were first made known to modern scholars by Silvestre de Sacy, who discovered them in an Arabic manuscript from the library of the ancient abbey of St.-Germain-des-Pres. The system has nine characters, but no zero. A dot above a character indicates tens, two dots hundreds, and so on, 5 meaning 50, and 5 meaning 5000.

In form these gober numerals resemble our own much more closely than the Arab numerals do. They varied more or less, but were substantially as follows.

This theory of the very early introduction of the numerals into Europe fails in several points. In the first place the early Western forms are not known; in the second place some early Eastern forms are like the gober, as is seen in the third line, where the forms are from a manuscript written at Shiraz about 970 A.D., and in which some western Arabic forms are also used.

The Boethius Question

First, who was Boethius,—*Divus* Boethius as he was called in the Middle Ages? **Anicius Manlius Severinus Boethius** was born at Rome c. 475. He was a member of the distinguished family of the Anicii, which had for some time before his birth been Christian. Early left an orphan, the tradition is that he was taken to Athens at about the age of ten, and that he remained there eighteen years. He married Rusticiana, daughter of the senator Symmachus, and this union of two such powerful families allowed him to move in the highest circles. Standing strictly for the right, and against all iniquity at court, he became the object of hatred on the part of all the unscrupulous element near the throne, and his bold defense of the ex-consul Albinus, unjustly accused of treason, led to his imprisonment at Pavia and his execution in 524.

Not many generations after his death, the period being one in which historical criticism was at its lowest ebb, the church found it profitable to look upon his execution as a martyrdom. He was accordingly looked upon as a saint,¹⁰ his bones were enshrined,¹¹ and as a natural consequence his books were among the classics in the church schools for a thousand years. It is pathetic, however, to think of the medieval student trying to extract mental nourishment from a work so abstract, so meaningless, so unnecessarily complicated, as the arithmetic of Boethius.

He was looked upon by his contemporaries and immediate successors as a master, for Cassiodorus (c. 490–c. 585 A.D.) says to him: “Through your translations the music of Pythagoras and the astronomy of Ptolemy are read by those of Italy, and the arithmetic of Nicomachus and the geometry of Euclid are known to those of the West.” Founder of the medieval scholasticism, distinguishing the trivium and quadrivium, writing the only classics of his time, Gibbon well called him “the last of the Romans whom Cato or Tully could have acknowledged for their countryman.”

The second question relating to Boethius is this: Could he possibly have known the Hindu numerals? In view of the relations that will be shown to have existed between the East and the West, there can only be an affirmative answer to this question. The numerals had existed, without the zero, for several centuries; they had been well known in India; there had been a continued interchange of thought between the East and West; and warriors, ambassadors, scholars, and the restless trader, all had gone back and forth, by land or more frequently by sea, between the Mediterranean lands and the centers of Indian commerce and culture. Boethius could very well have learned one or more forms of Hindu numerals from some traveler or merchant.

To justify this statement it is necessary to speak more fully of these relations between the Far East and Europe. It is true that we have no records of the interchange of learning, in any large way, between eastern Asia and central Europe in the century preceding the time of Boethius. But it is one of the mistakes of scholars to believe that they are the sole transmitters of knowledge.

As a matter of fact there is abundant reason for believing that Hindu numerals would naturally have been known to the Arabs, and even along every trade route to the remote west, long before the zero entered to make their place-value possible, and that the characters, the methods of calculating, the improvements that took place from time to time, the zero when it appeared, and the customs as to solving business problems, would all have been made known from generation to generation along these same trade routes from the Orient to the Occident.

It must always be kept in mind that it was to the tradesman and the wandering scholar that the spread of such learning was due, rather than to the school man. Indeed, Avicenna (980–1037 a.d.) in a short biography of himself relates that when his people were living at Bokhara his father sent him to the house of a grocer to learn the Hindu art of reckoning, in which this grocer (oil dealer, possibly) was expert. Leonardo of Pisa, too, had a similar training.

¹⁰Hence the *Divus* in his name.

¹¹Thus Dante, speaking of his burial place in the monastery of St. Pietro in Ciel d'Oro, at Pavia, says: “The saintly soul, that shows / The world’s deceitfulness, to all who hear him, / Is, with the sight of all the good that is, / Blest there. The limbs, whence it was driven, lie / Down in Cieldauro; and from martyrdom / And exile came it here.”—*Paradiso*, Canto X.

Even in very remote times, before the Hindu numerals were sculptured in the cave of Nana Ghat, there were trade relations between Arabia and India. Indeed, long before the Aryans went to India the great Turanian race had spread its civilization from the Mediterranean to the Indus. At a much later period the Arabs were the intermediaries between Egypt and Syria on the west, and the farther Orient. In the sixth century B.C., Hecataeus, the father of geography, was acquainted not only with the Mediterranean lands but with the countries as far as the Indus.

A century after Scylax, Herodotus showed considerable knowledge of India, speaking of its cotton and its gold, telling how Sesostris fitted out ships to sail to that country, and mentioning the routes to the east. These routes were generally by the Red Sea, and had been followed by the Phoenicians and the Sabaeans, and later were taken by the Greeks and Romans.

In the fourth century B.C. the West and East came into very close relations. As early as 330, Pytheas of Massilia (Marseilles) had explored as far north as the northern end of the British Isles and the coasts of the German Sea, while Macedon, in close touch with southern France, was also sending her armies under Alexander through Afghanistan as far east as the Punjab. Pliny tells us that Alexander the Great employed surveyors to measure the roads of India; and one of the great highways is described by Megasthenes, who in 295 B.C., as the ambassador of Seleucus, resided at Pataliputra, the present Patna.

The Hindus also learned the art of coining from the Greeks, or possibly from the Chinese, and the stores of Greco-Hindu coins still found in northern India are a constant source of historical information. The Ramayana speaks of merchants traveling in great caravans and embarking by sea for foreign lands. Ceylon traded with Malacca and Siam, and Java was colonized by Hindu traders, so that mercantile knowledge was being spread about the Indies during all the formative period of the numerals.

As to the Arabs, it is a mistake to feel that their activities began with Mohammed. Commerce had always been held in honor by them, and the Qoreish had annually for many generations sent caravans bearing the spices and textiles of Yemen to the shores of the Mediterranean. In the fifth century they traded by sea with India and even with China, and Hira was an emporium for the wares of the East, so that any numeral system of any part of the trading world could hardly have remained isolated.

Long before the warlike activity of the Arabs, Alexandria had become the great market-place of the world. From this center caravans traversed Arabia to Hadramaut, where they met ships from India. Others went north to Damascus, while still others made their way along the southern shores of the Mediterranean. Hindus were found among the merchants who frequented the bazaars of Alexandria, and Brahmins were reported even in Byzantium.

Such is a very brief resume of the evidence showing that the numerals of the Punjab and of other parts of India as well, and indeed those of China and farther Persia, of Ceylon and the Malay peninsula, might well have been known to the merchants of Alexandria, and even to those of any other seaport of the Mediterranean, in the time of Boethius. The Brahmi numerals would not have attracted the attention of scholars, for they had no zero so far as we know, and therefore they were no better and no worse than those of dozens of other systems.

In answer therefore to the second question, Could Boethius have known the Hindu numerals? the reply must be, without the slightest doubt, that he could easily have known them, and that it would have been strange if a man of his inquiring mind did not pick up many curious bits of information of this kind even though he never thought of making use of them.

As to the fourth question, Did Boethius probably know the numerals? It seems to be a fair conclusion, according to our present evidence, that (1) Boethius might very easily have known these numerals without the zero, but, (2) there is no reliable evidence that he did know them. And just as Boethius might have come in contact with them, so any other inquiring mind might have done so either in his time or at any time before they definitely appeared in the tenth century. These centuries, five in number, represented the darkest of the Dark Ages, and even if these numerals were occasionally met and studied, no trace of them would be likely to show itself in the literature of the period. As a matter of fact, it was not until the ninth or tenth century that there is any tangible evidence of their presence in Christendom.

6 The Development of the Numerals Among the Arabs

When the Arabs, carried on the wave of religious enthusiasm, had spread their conquests from the Atlantic to the Indus, they found everywhere the use of the Greek alphabetic numerals. In Syria they adopted the Greek system, in Egypt the Coptic, and in Persia the system used by the Persians. These systems were, however, of little value for mercantile purposes, and the Arabs early saw the advantages of the Hindu system which had reached them through the translations of the astronomical works of Sindhind.

The spread of the numerals among the Arabs was slow. In the East, under the Abbasid caliphs of Bagdad, the Hindu system found ready acceptance among scholars, but the common people and the merchants continued to use their old systems. It was not until the eleventh century that the Hindu numerals became common in Arabic works, and even then they were used chiefly by mathematicians and astronomers.

The earliest known Arabic work containing the Hindu numerals is the arithmetic of Al-Khowarazmi, already mentioned. This work, which bore the title *Algoritmi de numero Indorum*, was translated into Latin, probably by Adelhard of Bath, in the twelfth century. The Arabic original is lost, but several Latin versions exist, and from these we know that Al-Khowarazmi gave the nine Hindu numerals and explained their use in addition, subtraction, multiplication, and division.

The numerals as used by the Arabs underwent certain modifications. The forms used in the Eastern part of the Arab empire, centering in Bagdad, were different from those used in the Western part, centering in Cairo and later in Spain. The Eastern forms were called *Indien* (Indian), while the Western forms were called *gobar* (dust). The gobar forms, which have already been described in connection with the Boethius question, were the ones used in Spain and Northern Africa, and from these the European forms were derived.

Among the Arab writers who used and spread the Hindu numerals may be mentioned Al-Kindi (800–870), Al-Sufi (died 986), Al-Biruni (973–1048), and Al-Hassar (c. 1150). Al-Kindi wrote several works on arithmetic and the use of the Indian method of reckoning. Al-Sufi wrote a large work on arithmetic. Al-Biruni, as already mentioned, was a great traveler and scholar who used the Hindu numerals extensively in his astronomical and mathematical works. Al-Hassar wrote a work on arithmetic in which the second chapter was “On the dust figures.”

The Arabic system of numeration, like the Hindu, was based on the principle of place value. The Arabs adopted the zero, which they called *as-sifr* or *sifr*, and used it in the same way as the Hindus. They also adopted the Hindu methods of calculation, including the rules for the fundamental operations and the extraction of roots. These methods were known as *hisab al-hind*, the Indian reckoning, and *hisab al-gobar*, the dust reckoning.

The spread of the numerals among the Arabs was thus a gradual process, extending over several centuries. It was aided by the great intellectual activity that characterized the Arab world from the ninth to the twelfth century, and by the fact that the Arabs were in contact with both the East and the West, serving as intermediaries in the transmission of knowledge.

7 The Definite Introduction of the Numerals into Europe

The definite introduction of the Hindu-Arabic numerals into Europe is usually attributed to Leonardo of Pisa, known as Fibonacci, who wrote his famous *Liber Abbaci* in 1202. Leonardo had traveled extensively in the East, visiting Egypt, Syria, Greece, and Sicily, and had learned the Hindu methods of reckoning from Arab teachers. In his book he explained the use of the nine Indian figures and the zero, and showed how calculations could be performed with them.

The *Liber Abbaci* was not, however, the first work in Europe to mention the Hindu numerals. As we have seen, there is evidence that the numerals were known in Spain and possibly in other parts

of Europe as early as the tenth century. But Leonardo's work was the first to give a systematic and comprehensive treatment of the new system, and it was through his influence that the numerals were gradually adopted in Italy and thence spread to other parts of Europe.

The reception of the numerals in Europe was slow. The medieval universities, with their devotion to the classics, were at first hostile to the new system, which was considered a novelty of doubtful value. The Roman numerals were deeply entrenched in European culture, and it required several centuries before the Hindu-Arabic numerals were generally accepted.

An important figure in the early history of the numerals in Europe was Gerbert, who became Pope Sylvester II in 999. Gerbert had studied in Spain and had learned the numerals there. He introduced them into the schools of France, but in a modified form, using apices or counters on an abacus rather than the written numerals. This system, known as the Gerbert abacus, was used in European schools for several centuries.

The real progress in the adoption of the numerals came in the thirteenth and fourteenth centuries, when Italian merchants began to use them in their business transactions. The commercial needs of the Italian city-states, with their extensive trade with the East, made the new system highly attractive. The Italian mathematicians, following Leonardo of Pisa, wrote numerous works on arithmetic in which the Hindu numerals were used and explained.

Among the early European works on arithmetic using the Hindu numerals may be mentioned the *Carmen de Algorismo* of Alexander de Villa Dei (c. 1240), the *Algorismus* of Johannes de Sacrobosco (c. 1250), and the various Italian arithmetics of the fourteenth century. These works helped to spread the knowledge of the numerals and to familiarize European scholars and merchants with the new system.

The spread of the numerals from Italy to the rest of Europe was a gradual process. France and Germany adopted them in the fifteenth century, and England in the sixteenth. Even then, the Roman numerals continued to be used for many purposes, and it was not until the eighteenth century that the Hindu-Arabic numerals became completely dominant in Europe.

8 The Spread of the Numerals in Europe

After the numerals had been introduced into Italy, they gradually spread to the other countries of Europe. The process was slow, for the Roman numerals were deeply rooted in European culture, and the new system encountered strong resistance from conservative scholars and ecclesiastics.

In France, the Hindu numerals were known as early as the thirteenth century, but they did not come into general use until the fifteenth. The first French work to use them extensively was the arithmetic of Jehan Adam, written in 1475. In the sixteenth century, French mathematicians such as Chuquet (1484) and Trenchant (1566) helped to popularize the new system.

In Germany, the numerals were introduced in the fifteenth century, but they met with strong opposition. The German merchants preferred to use the abacus, and the scholars were attached to the Roman system. It was not until the publication of the arithmetic of Kobel in 1514 that the Hindu numerals began to gain ground in Germany.

In England, the numerals were known in the fourteenth century, but they came into general use only in the seventeenth century. The first English work on arithmetic, *The Ground of Artes* by Robert Recorde (1542), used the Hindu numerals, but it was long before they replaced the Roman system entirely.

In Spain and Portugal, where the numerals had been known since the Moorish conquest, they were adopted earlier than in the rest of Europe. But even there the transition was gradual, and the Roman numerals continued to be used for official and ecclesiastical purposes for many centuries.

The spread of the numerals in Europe was thus a slow and gradual process, extending over several centuries. It was aided by the invention of printing, which made it possible to publish arithmetic textbooks in large numbers, and by the growth of commerce, which created a demand for a more

efficient system of numeration. By the eighteenth century, the Hindu-Arabic numerals had become the standard system throughout Europe, and they have since spread to all parts of the world.

Conclusion

In conclusion, we may say that the history of the Hindu-Arabic numerals is a long and complex one, stretching over many centuries and involving many different civilizations. From their earliest beginnings in India, through their development by the Arabs, to their final adoption in Europe, these numerals have had a remarkable career. They are one of the most important inventions in the history of civilization, and their influence on science, commerce, and daily life can hardly be overestimated.

The study of their history reveals the deep connections that have always existed between the peoples of the East and the West, and shows how knowledge has been transmitted from one civilization to another along the great trade routes that have linked the world since ancient times. It is a story that deserves to be known by all who use these symbols, which have become so familiar that we are apt to forget how much labor and ingenuity went into their creation.

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